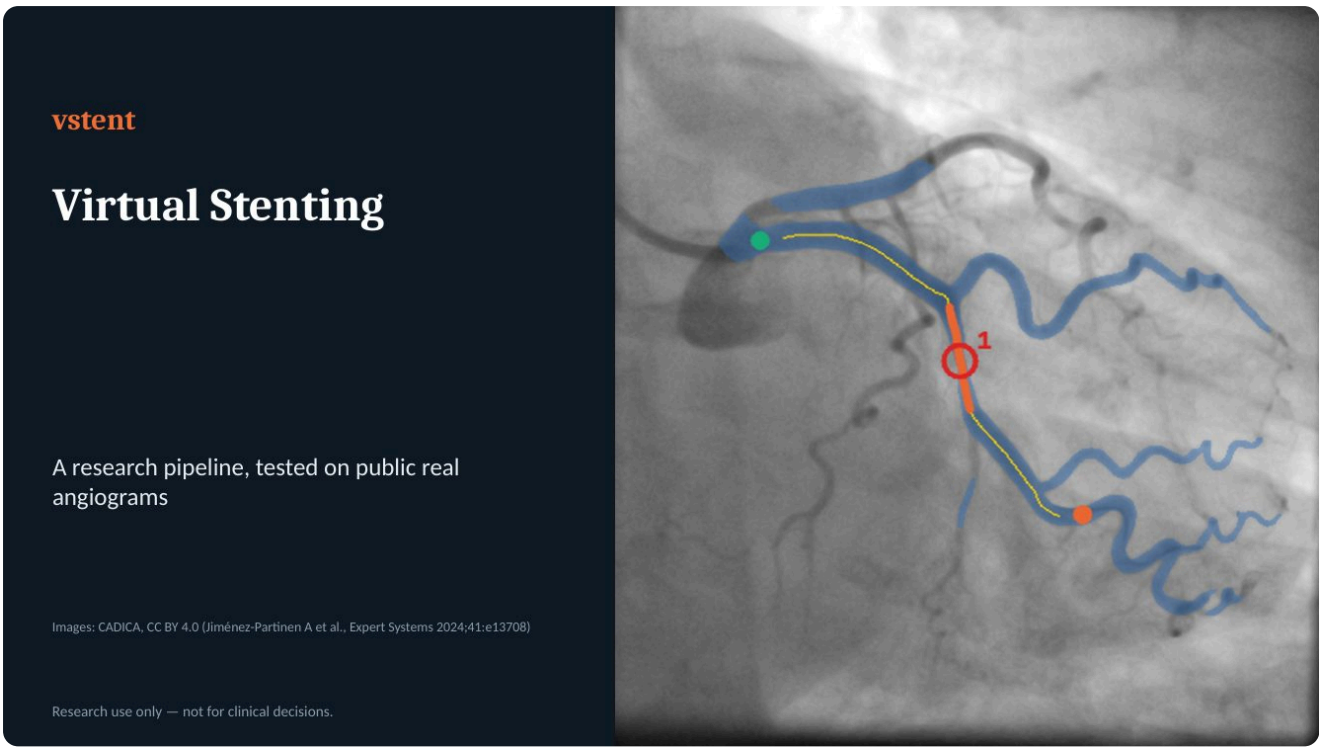
Speaker notes

Virtual Stenting

Notes for **vstent_research_Final.pptx**

Main talk: slides 1 to 12, about 15 minutes.
Slide 13: backup index. Stop here at the end of the talk.
Slides 14 to 39: backup, for questions only.

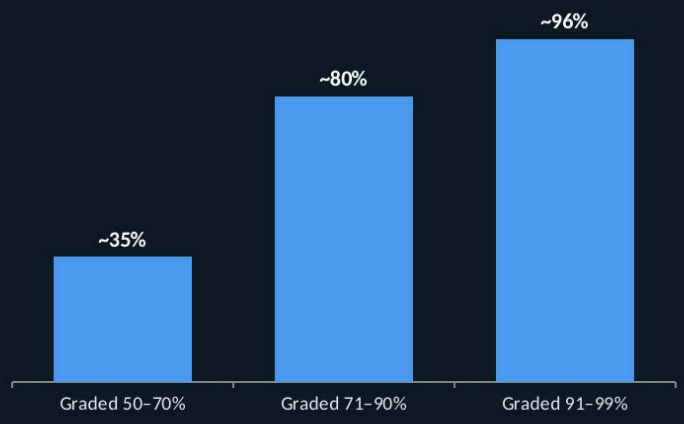
Research use only. Not for clinical decisions.



Thank you. This talk asks one question: can we get what a pressure wire tells us from an angiogram that is already recorded? The picture is a real patient from a public dataset, processed by our pipeline. By the end you will know what every colour means. This is research. It has not been compared with a pressure wire, and it must not be used to treat patients.

Seeing a narrowing is not knowing whether it matters

Lesions with wire FFR ≤ 0.80, by visual grade



FAME, Tonino et al., JACC 2010. FFR ≤ 0.80: ESC 2019 guidelines

Research question

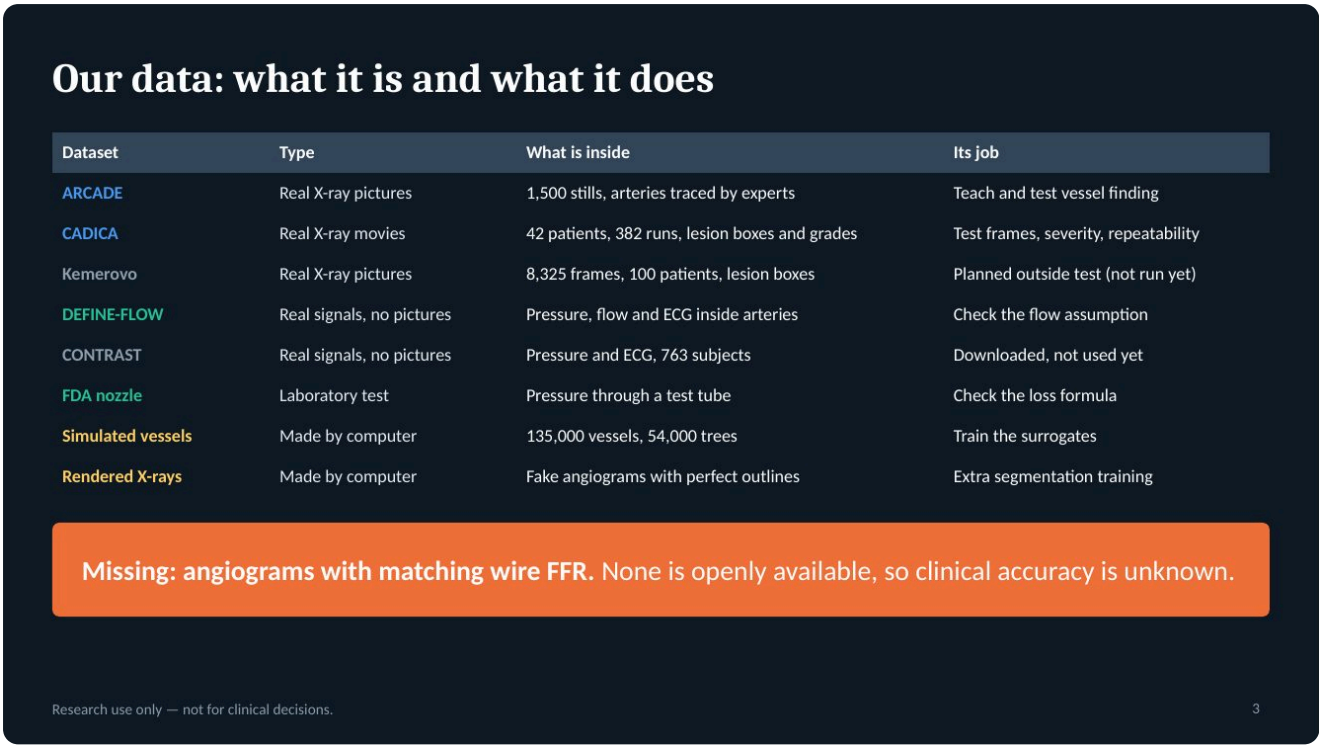
Can FFR be computed from the cine angiogram alone?

FFR = pressure beyond the narrowing ÷ aortic pressure at maximal flow

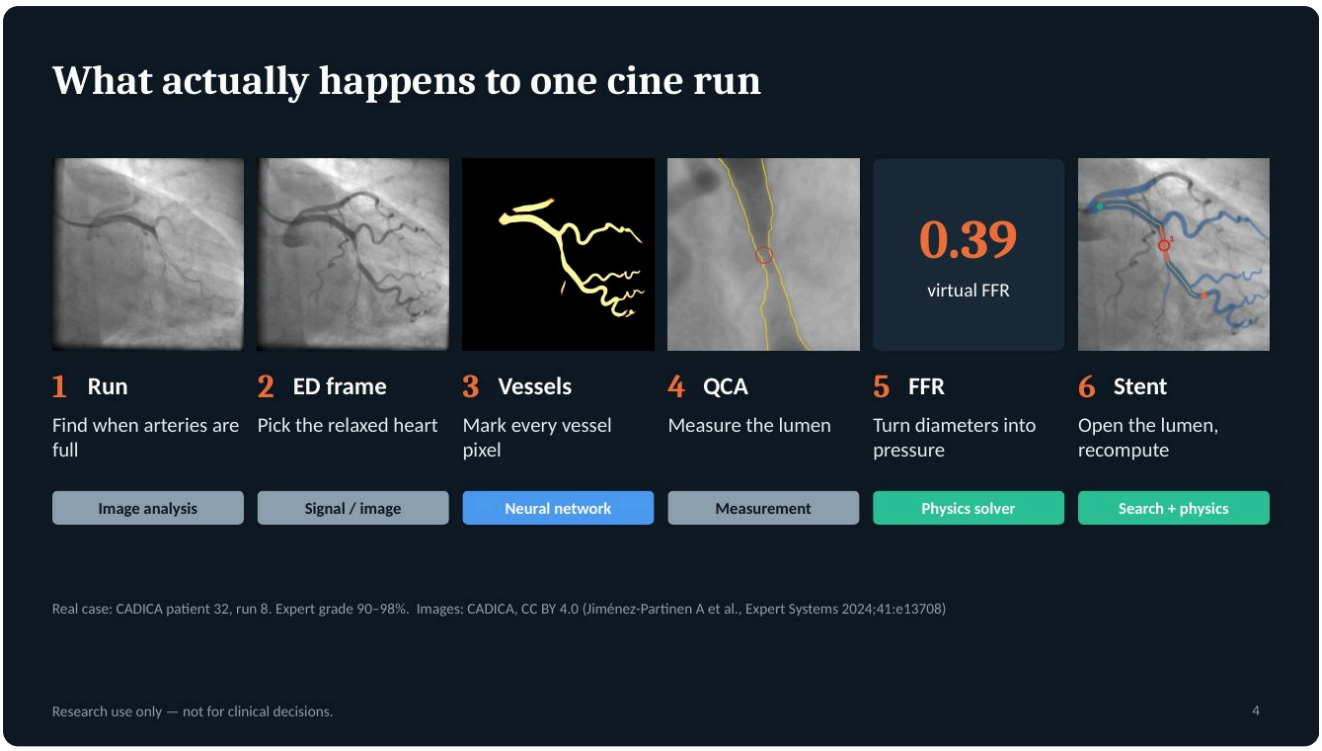
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2

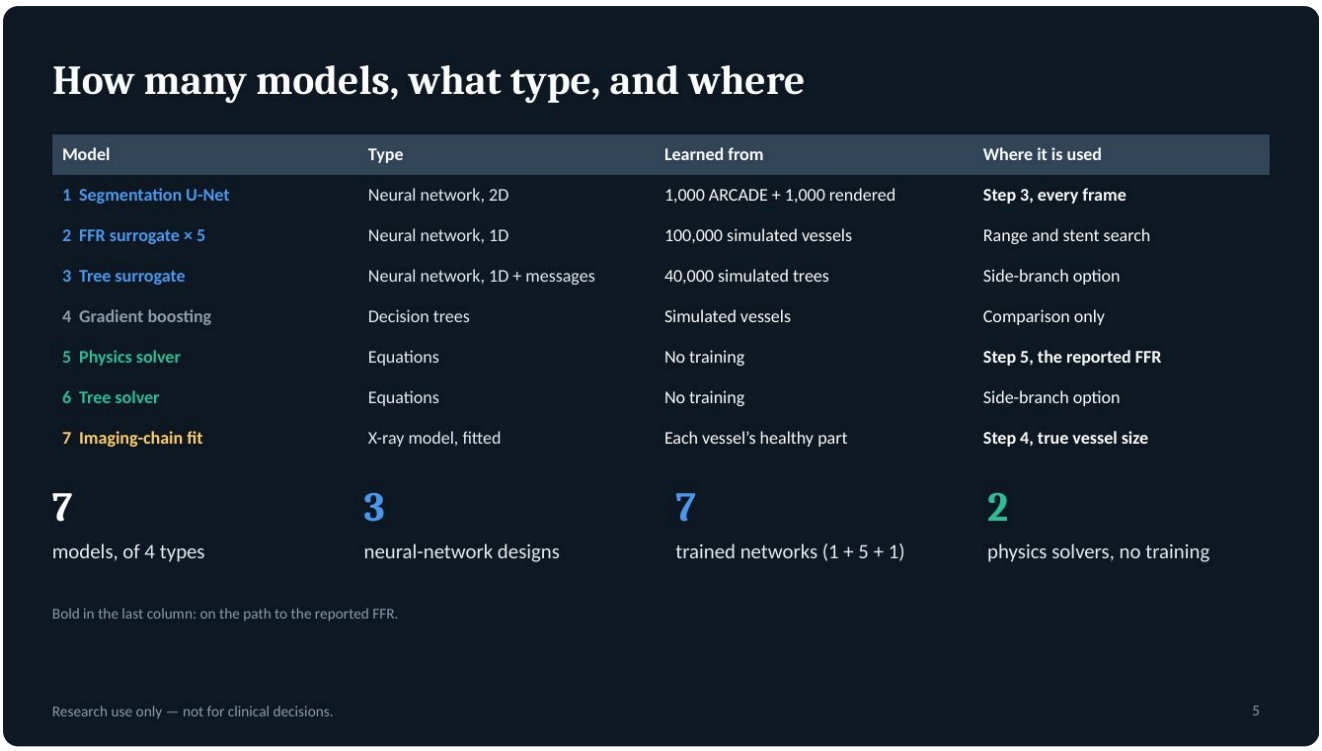
Doctors decide on stents by asking whether a narrowing limits flow, not just how tight it looks. FFR measures that: pressure beyond the narrowing divided by aortic pressure at maximal flow. At or below 0.80 it counts as significant. The eye is a poor judge. In FAME, only about 35 percent of lesions graded 50 to 70 percent were significant. The wire answers this, but it takes time, costs money and needs adenosine. So our question is: can the film alone give the answer?



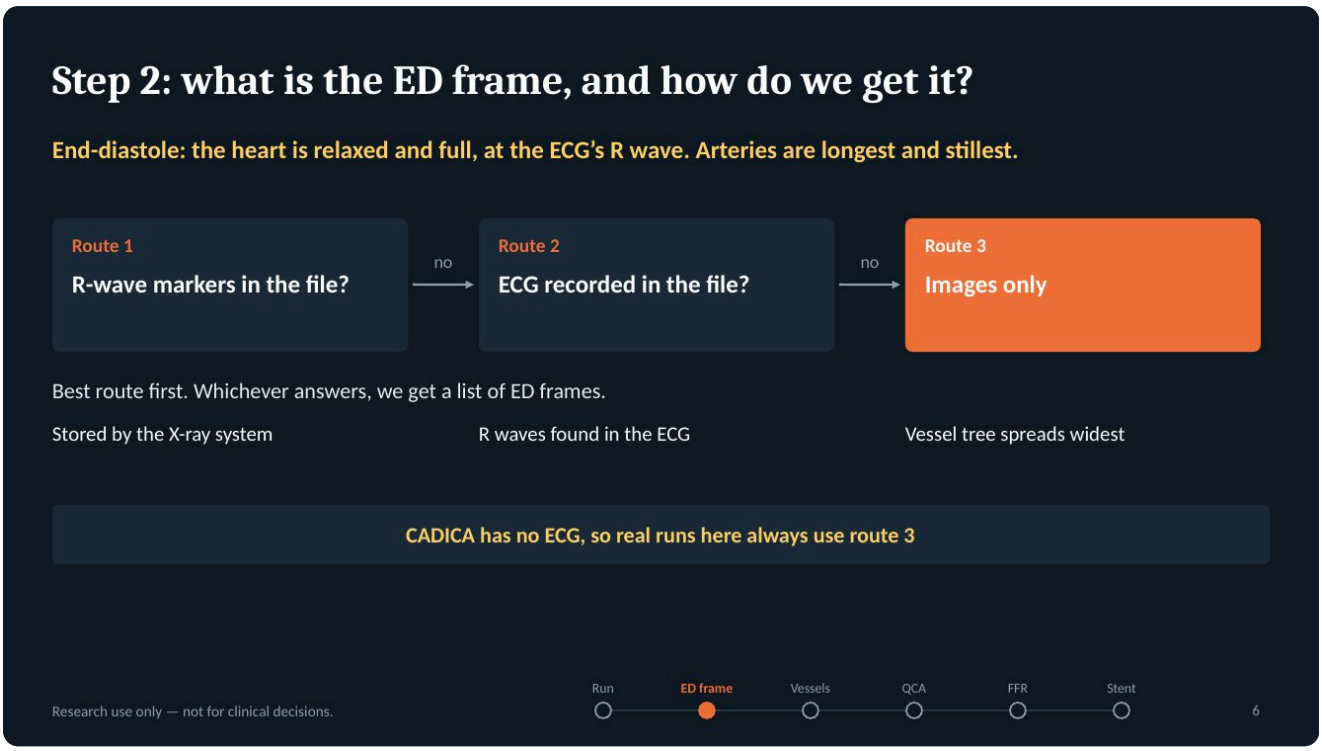
Before any model, here is our data. Think of data as the textbooks and exams our models learn from. We have three kinds. Real X-rays. Real measurements from inside arteries. And data a computer made, where we know the exact answer. Each row says what it is, what is inside, and what job it does. The last line is the most important. Nobody has openly published angiograms with matching wire FFR. We searched again on 2 November and found none. That is why clinical accuracy is still unknown.



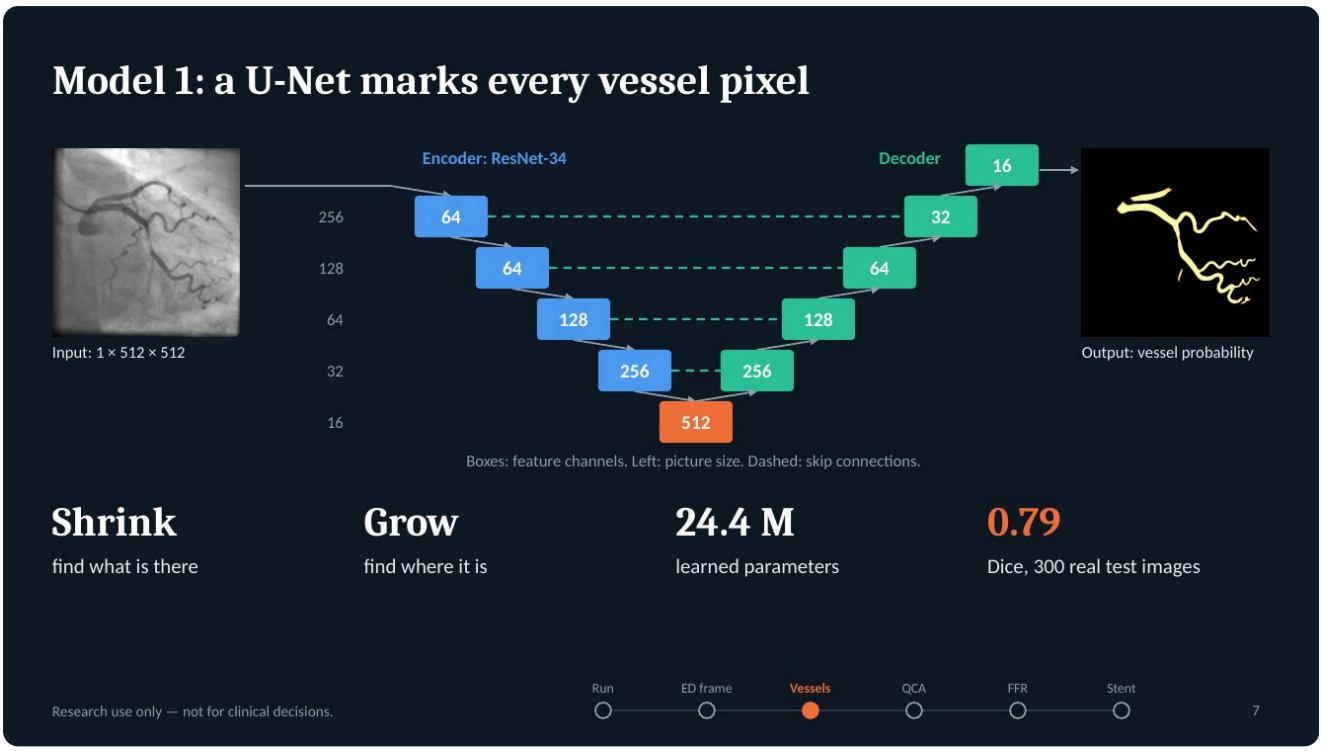
This is the whole method, on one real patient: CADICA patient 32, graded 90 to 98 percent by experts. Six steps. Read the film. Choose the frame. Find the vessels. Measure them. Compute pressure. Try a stent. Look at the tags underneath. Only one step needs a deep network to give the reported number. The rest is measurement and physics, so every number can be checked. Next, the models behind these steps.



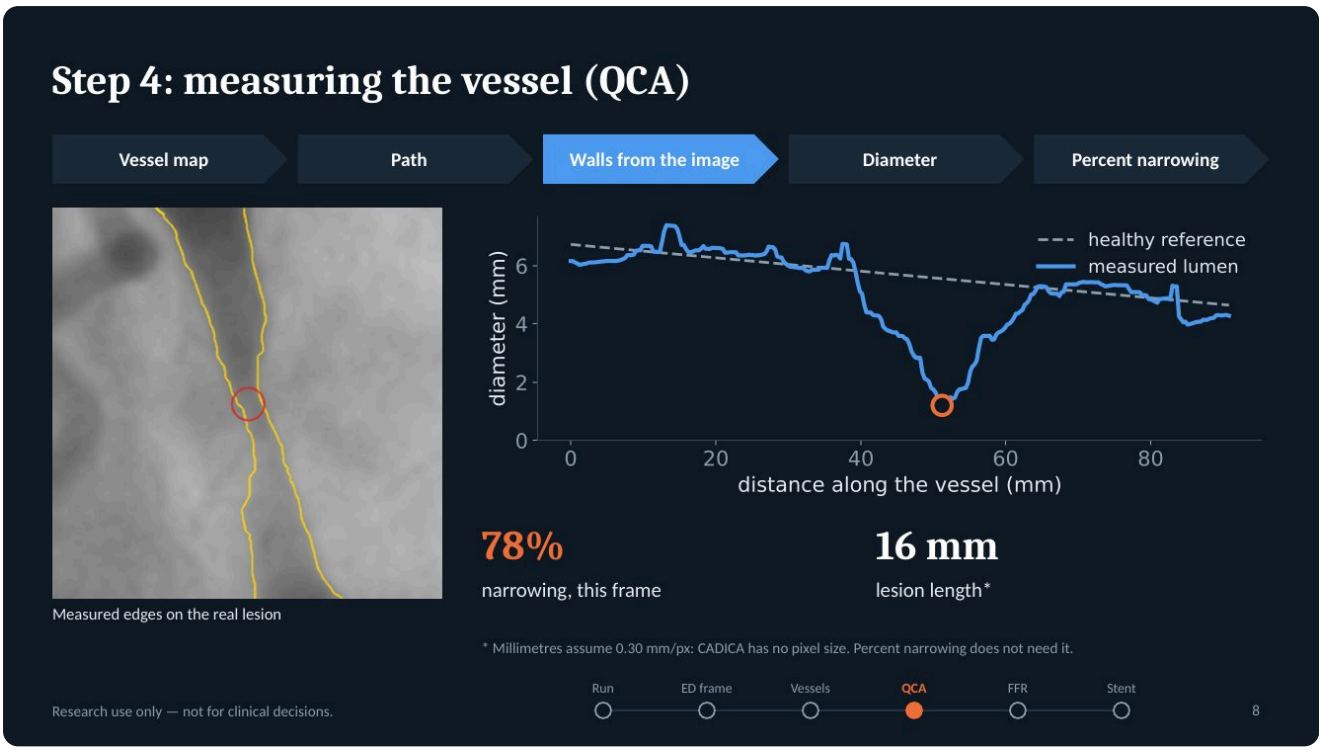
So how many models? Seven, of four types. Three are neural networks, and that is seven networks in total when you count copies: one segmentation network, five copies of the FFR surrogate, and one tree surrogate. One is a simple baseline, kept only for comparison. Two are physics solvers with no training at all. One is an X-ray model fitted to each vessel. Read the last column: only the U-Net, the imaging-chain fit and the physics solver produce the reported FFR. The others give the range, the side-branch option and the stent search.



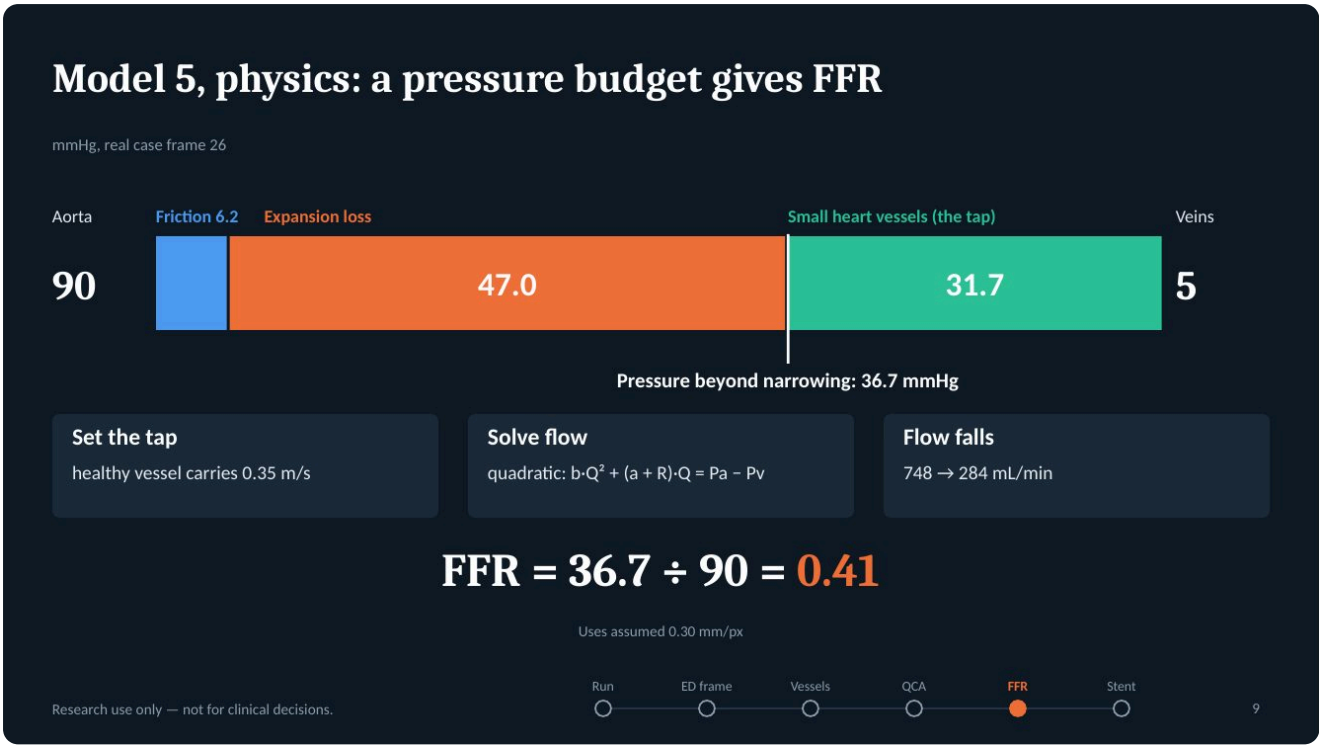
Step two is the end-diastolic frame. That is the moment the heart is relaxed and full, right at the R wave of the ECG. The arteries are longest and moving least, so it is the standard frame for measurement. The software tries three routes, best first. R-wave markers stored in the DICOM file. If none, R waves detected in a recorded ECG. If none, the images alone: the vessel tree spreads widest when the heart is full. We then pick the ED frame with the most vessel, measure there, and repeat on up to five filled frames for a median. One honest limit: CADICA has no ECG, so real runs here use route 3, and those frames are estimates.



Model one finds the vessels. It is a U-Net, named for its shape. On the left, the encoder shrinks the picture from 512 pixels to 16 while it learns more and more features, like going from edges to whole vessels. That part is a ResNet-34, a proven image network, started from ImageNet weights. On the right, the decoder grows it back to full size. The green dashed lines are skip connections: they pass fine detail straight across so thin vessels are not lost. The output is a probability for each pixel: is this vessel? It has 24.4 million parameters, and scores a Dice of 0.79 on 300 real test images.

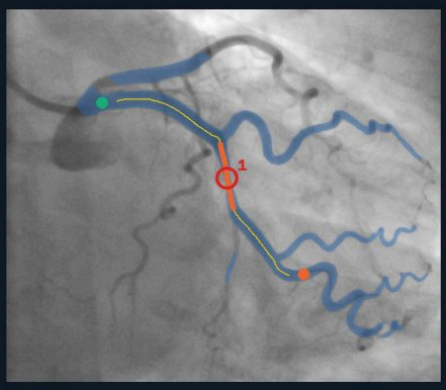


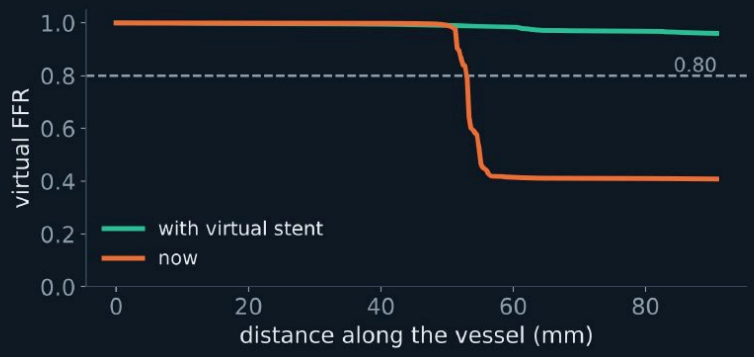
Step four measures the vessel. This is classical QCA, not AI. We thin the network's map to a centreline. At each point we walk across the vessel and find where brightness rises fastest. That gives the two walls, here in yellow. Importantly, the walls come from the X-ray image, because networks smooth over narrowings. The distance between the walls is the diameter, the blue line. The dashed line is the healthy reference. The circle is the tightest point: 78 percent narrowing on this frame. The millimetres assume 0.30 mm per pixel, because CADICA gives no pixel size. Percent narrowing does not depend on that.



Now the physics, and there is no AI here. A narrowing loses pressure in two ways: friction along the wall, and an expansion loss as blood jets out and swirls. Think of a water circuit. Pressure starts at 90 in the aorta and ends at 5 in the veins. The small heart vessels act like a tap, set so a healthy vessel carries 0.35 metres per second. A narrowing adds resistance, so flow falls: here to 284 millilitres per minute. The losses must add to 85: 6.2 friction, 47.0 expansion, 31.7 small vessels. Pressure beyond the narrowing is 36.7, and 36.7 over 90 is 0.41. That is FFR for this frame.

The real case: FFR 0.39, then 0.96 with a virtual stent





0.39
median of 4 frames

0.37–0.56
input range

4.00 × 22 mm
shortest stent to 0.90

90–98%
expert grade; we read 78%

Chosen because the pipeline reads it well. No wire value exists. Assumes 0.30 mm/px. Images: CADICA, CC BY 4.0 (Jiménez-Partinen A et al., Expert Systems 2024;41:e13708)

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10

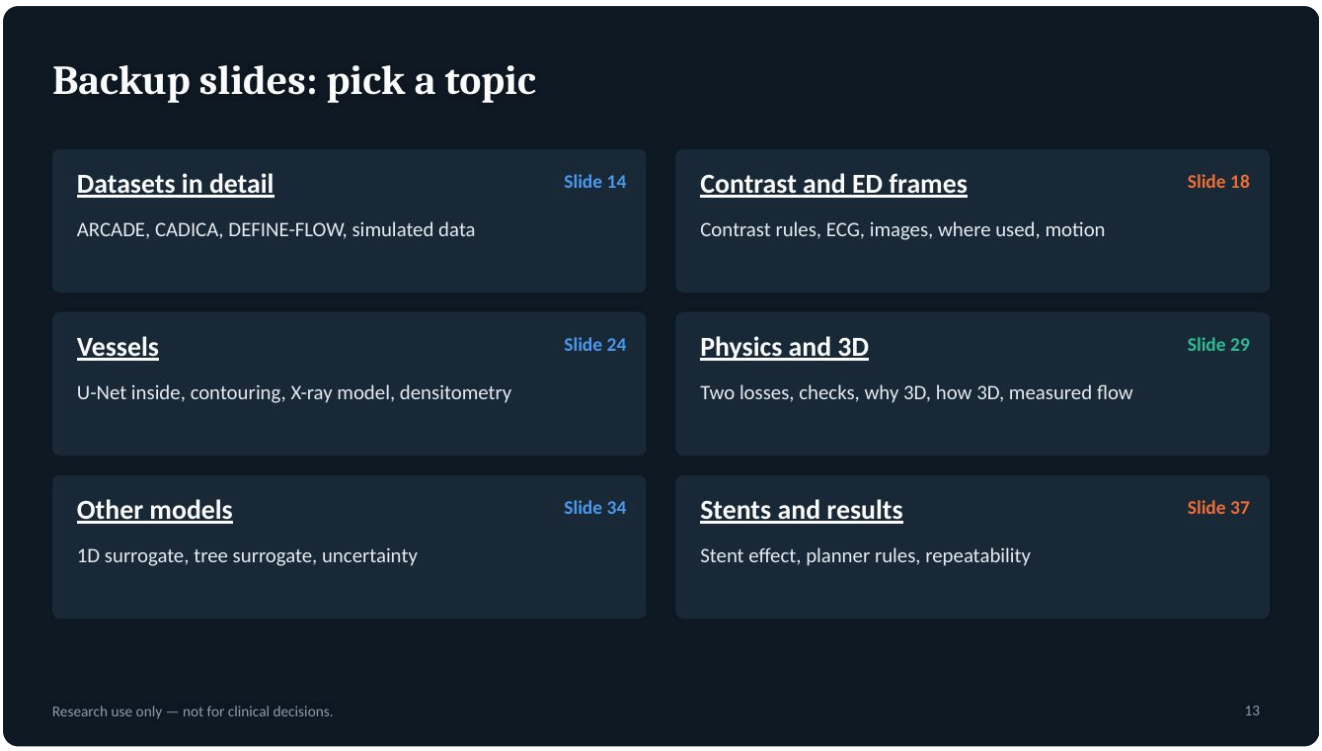
Here is the same patient end to end. FFR is 0.39, the median of four frames, with an input range of 0.37 to 0.56. A virtual stent opens the lumen and we rerun the physics. The planner searches a generic catalogue for the shortest stent reaching 0.90: here 4.00 by 22 millimetres, giving 0.96. Three warnings. I chose this case because the pipeline reads it well, so it is a demonstration, not evidence. There is no wire value, so we cannot say 0.39 is right. And the size assumes 0.30 mm per pixel. The group result is next, and it is less flattering.



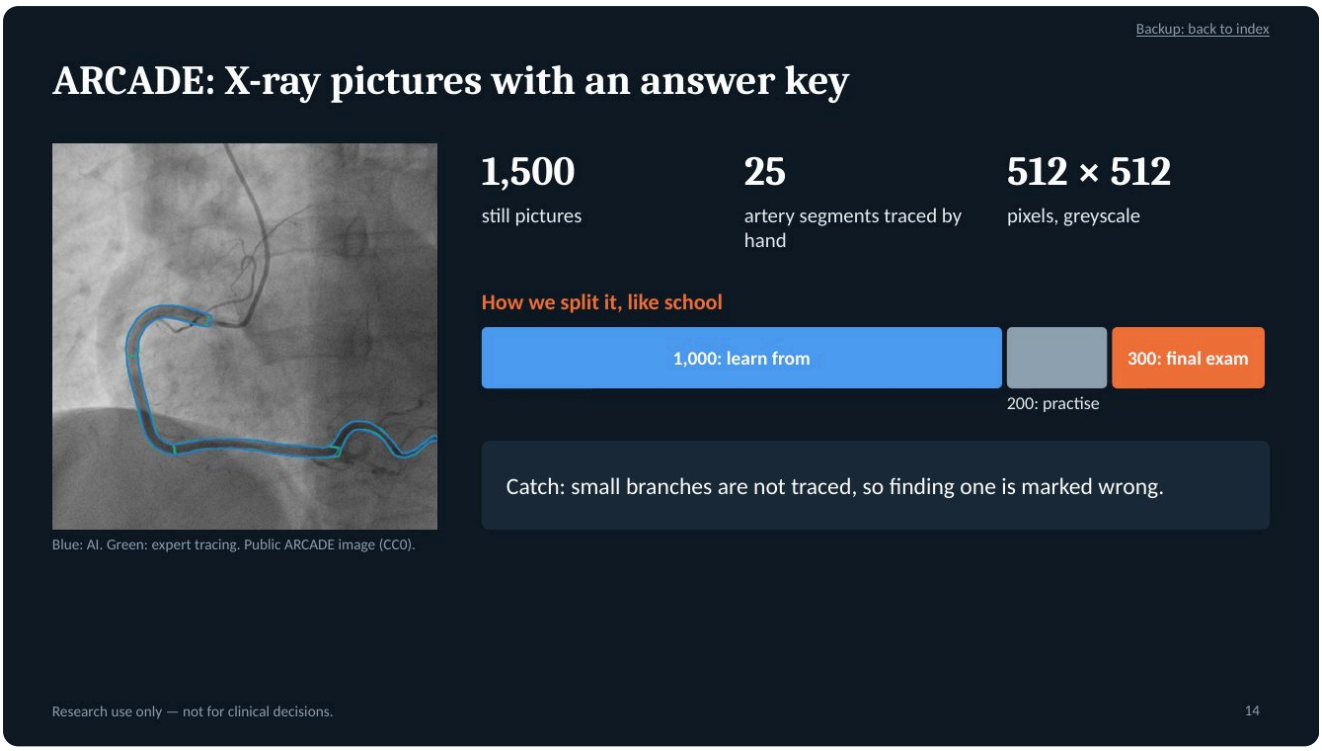
This is the main finding, and our own checks found it. Taking the median across frames helps repeatability: FFR variation fell from 18.7 percent to 5.1 on test patients, and 12.0 on development patients. But a median cannot remove a shared bias. In 82 test lesions graded 70 percent or more, none was measured at 70 percent or more. So the tool called only 16 and 23 percent of them significant, where wire studies find most significant. This case: experts say 90 to 98 percent, we said 0.96. A 90 percent narrowing in a 2 millimetre vessel is thinner than the X-ray blur. Densitometry raised agreement from 0.13 to 0.25, but severe lesions still read about 49 percent, so we did not adopt it.



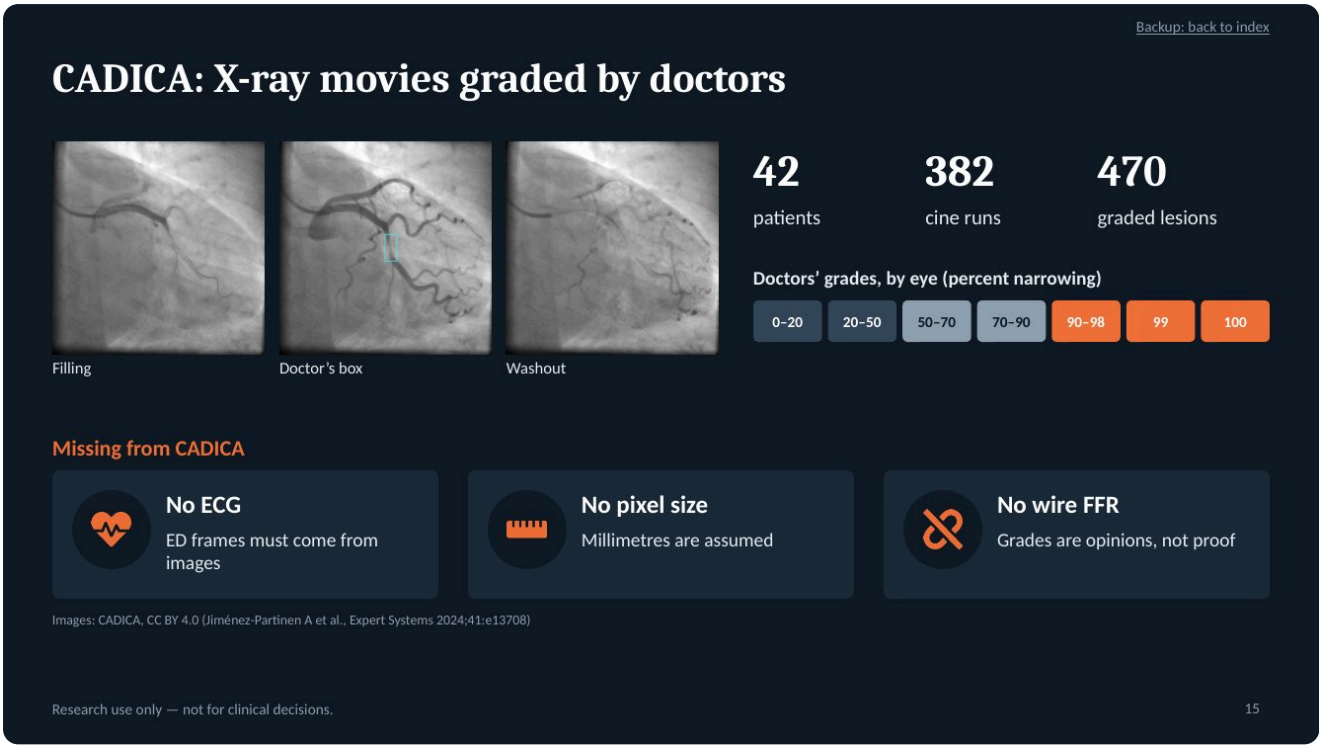
To finish. On real data we have shown vessel finding at Dice 0.79, frame choice that lands on well-filled frames, and better repeatability from a median. Not solved: tight lesions read too mild, the physics is simple, and inputs are assumed. All training data is public, the automatic path can pick the wrong vessel, and regulatory work is drafts only. The next step decides everything: a blinded comparison with the wire, predictions locked by checksum before wire values are opened, 30 to 50 vessels first, then a few hundred. For that we need clinical partners with FFR cases. Thank you.



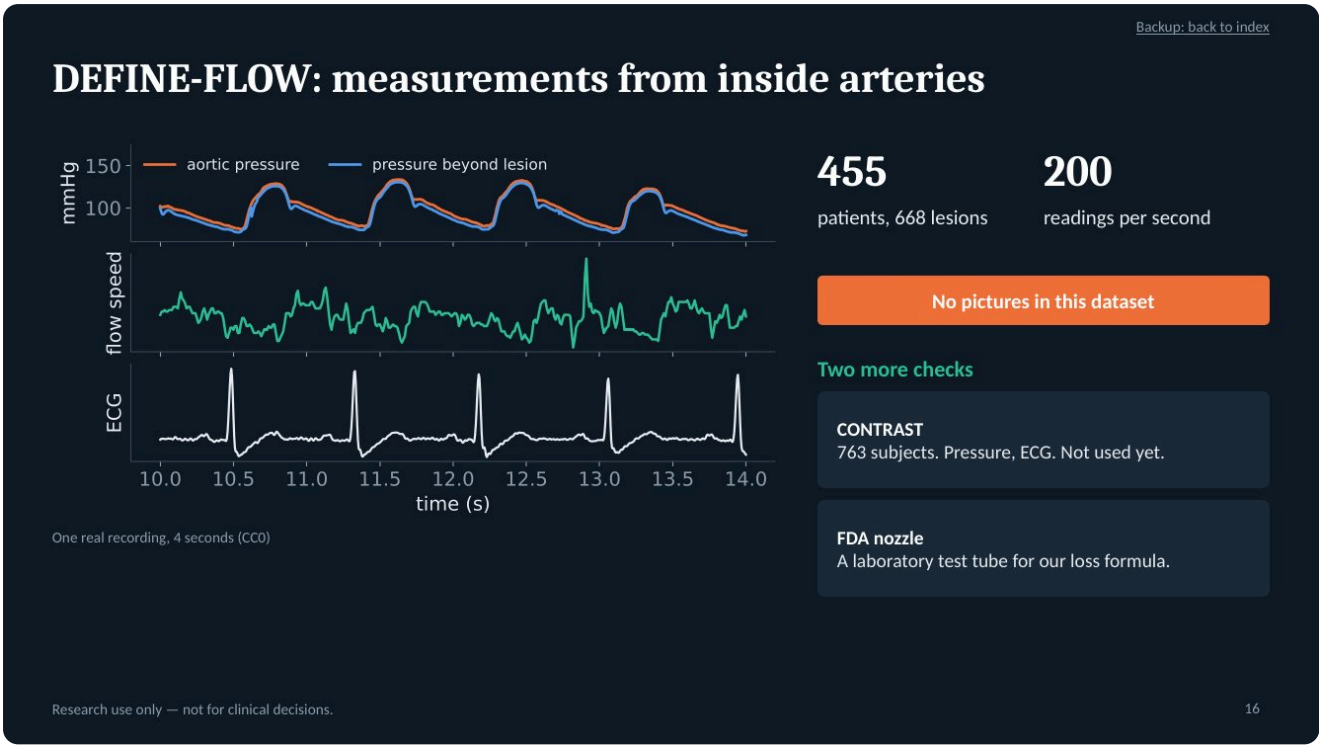
Stop here at the end of the talk. Click a topic to jump to its backup slides. Each backup slide has a link back to this page.



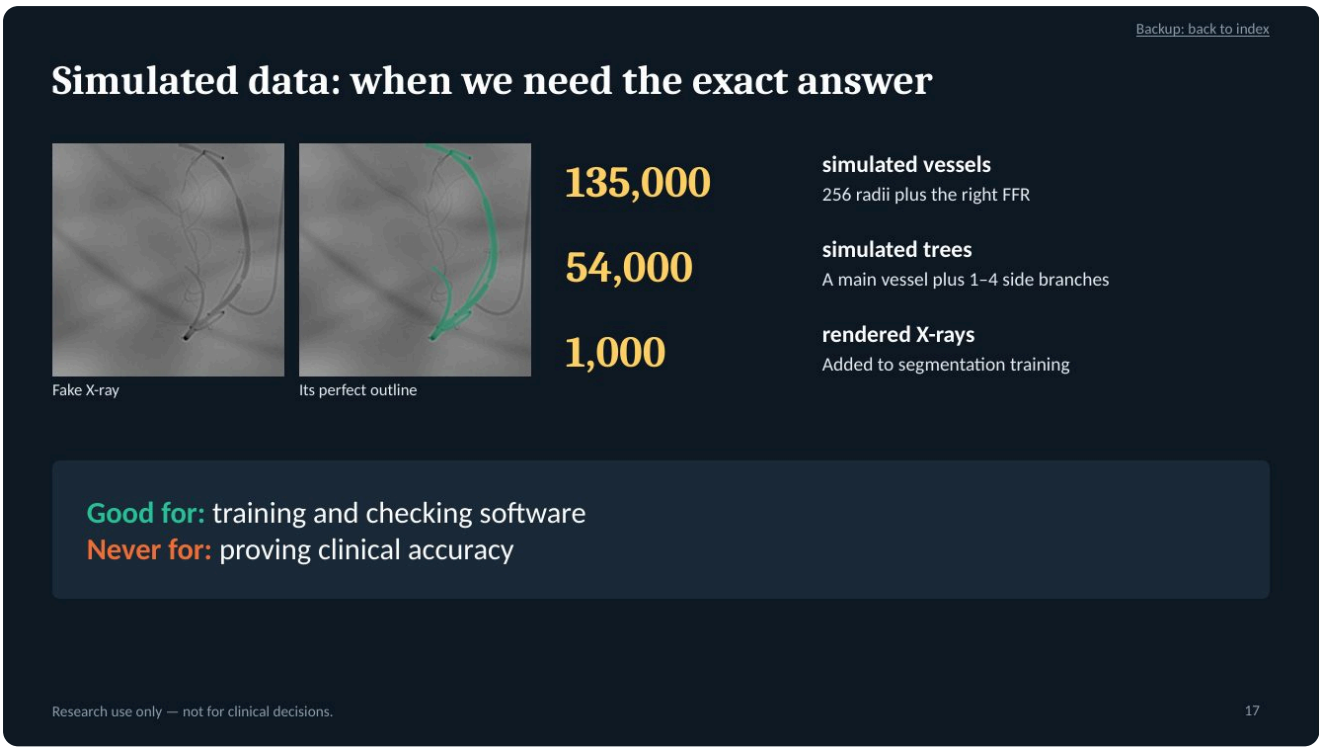
ARCADE is a set of still X-ray pictures. It is our textbook for finding vessels. There are 1,500 pictures, 512 by 512 pixels, in greyscale. Experts traced the arteries by hand, in 25 named segments. Those tracings are the answer key. We split it three ways, like school: 1,000 to learn from, 200 to practise on, 300 as a final exam the model sees once. One catch: experts did not trace every small branch, so a model that finds an untraced branch is marked wrong.



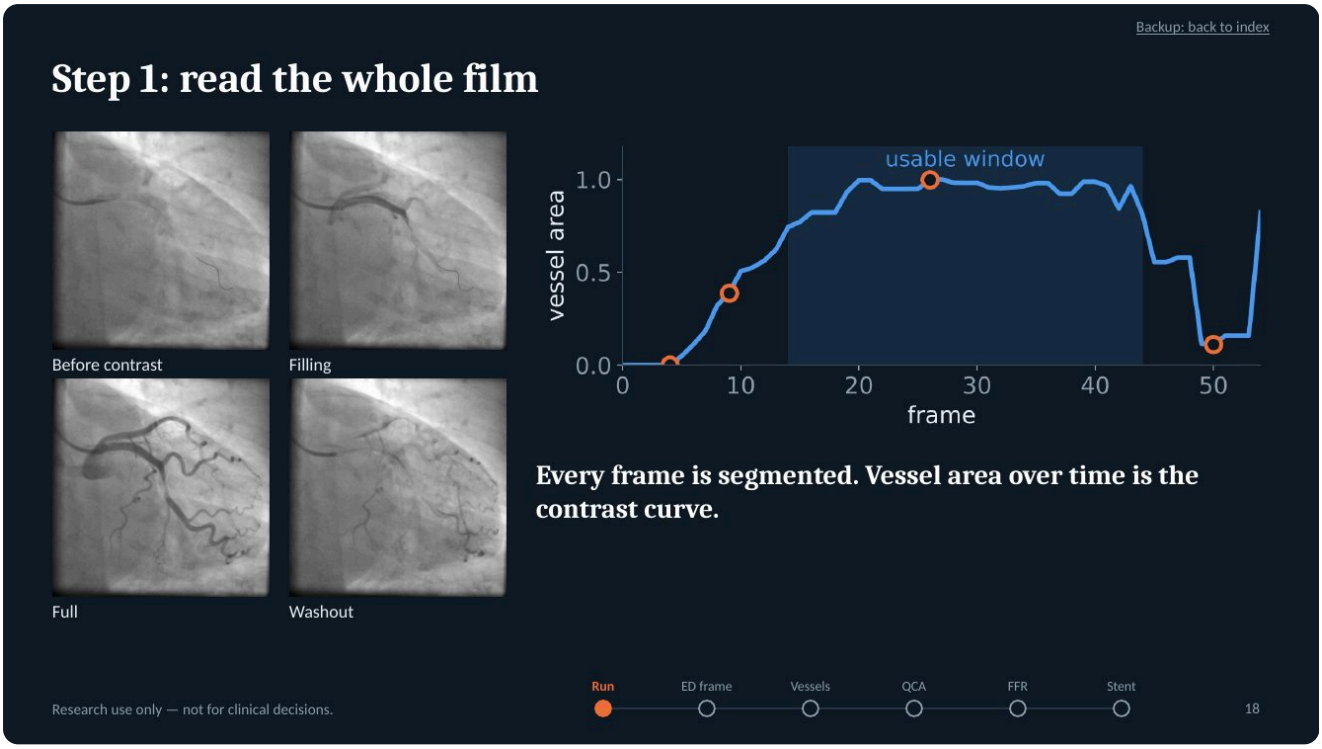
CADICA is different: these are full X-ray movies, 42 patients and 382 runs, at about 10 frames per second. Doctors drew boxes around narrowings, 6,161 boxes for 470 lesions, and graded each by eye from 0 to 100 percent. That is our closest thing to a clinical test. We split patients in half: 21 to develop on, 21 to test once. But three things are missing: no ECG, no pixel size and no wire FFR. So grades are opinions, not proof.



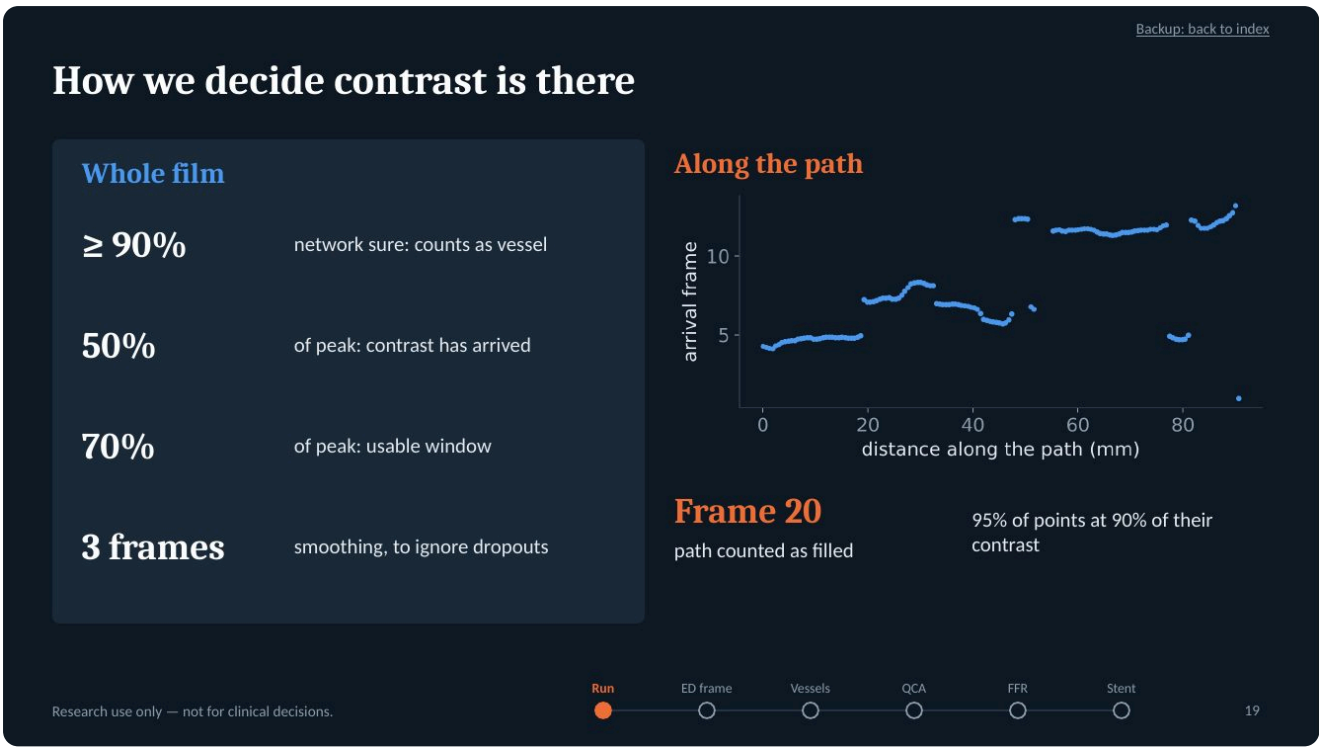
DEFINE-FLOW has no pictures at all. A sensor wire inside the artery recorded pressure, flow speed and ECG, 200 times a second. This is one real recording. The orange line is aortic pressure, the blue line is pressure beyond the narrowing. The gap between them is what FFR measures. The dataset covers 455 patients and 668 lesions. We used 455 lesions from 379 subjects to check one number in our physics. CONTRAST is similar, 763 subjects, downloaded but not used yet. The FDA nozzle is a laboratory test tube used to check our pressure-loss formula.



Why make data at all? Because with real patients we rarely know the exact answer, and a computer model does. We made three things. 135,000 simulated vessels: each is just a list of 256 radii, with the correct FFR from our solver. 54,000 simulated vessel trees with side branches. And rendered X-rays: fake angiograms with a perfect outline, and we added 1,000 of them to segmentation training. Simulated data is good for training and for checking software. It never proves clinical accuracy, so no clinical claim in this talk rests on it.

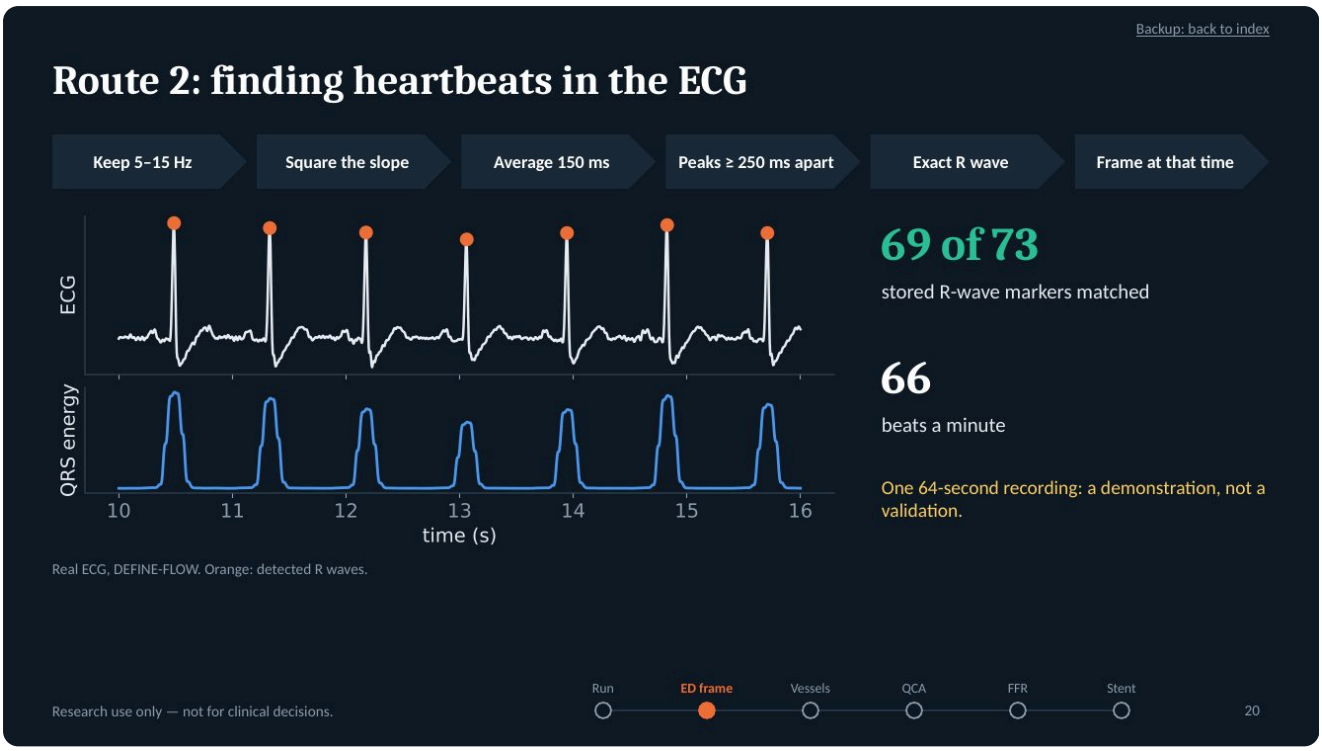


Step one reads the whole film. Each frame goes through the segmentation network and we count vessel pixels. That count over time is the contrast curve. It rises as dye arrives, stays high while arteries are full, and falls as the dye washes out. The four circles match the four frames on the left. The shaded part is the usable window: frames above 70 percent of the peak. Measure outside it and you measure a half-filled artery.

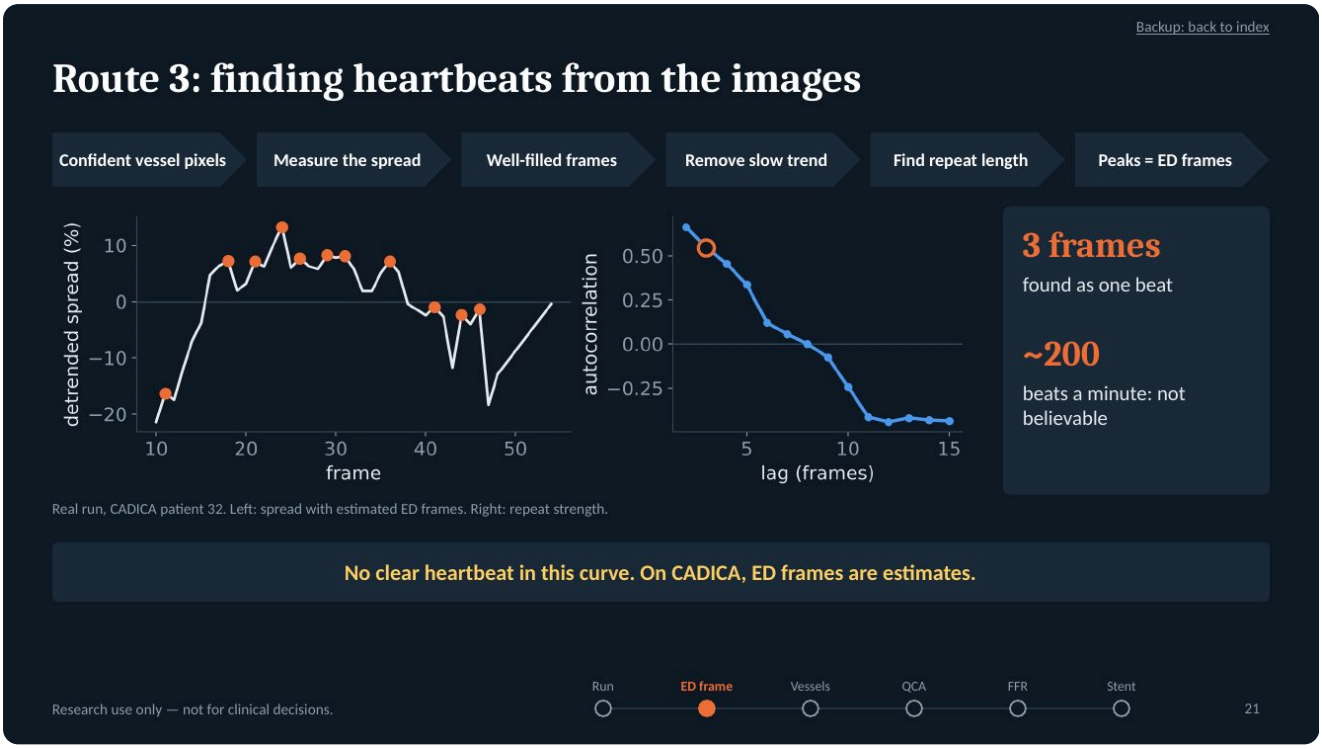


How do we decide contrast is there? At two levels. First, the whole film. A pixel counts as vessel only if the network is at least 90 percent sure. We count those pixels on each frame and smooth over 3 frames. Contrast has arrived when the count passes 50 percent of its peak, and the usable window is above 70 percent. Second, point by point along the path. At each point we take the darkest value within 6 pixels, following the heart, and compare it with that same point before the dye came. Arrival is when it reaches half of its final contrast. The chart shows dye reaching the far end about 8 frames after the start. The path counts as filled when 95 percent of its points reach 90 percent of their contrast: here, frame 20. Only frames from then on are used.

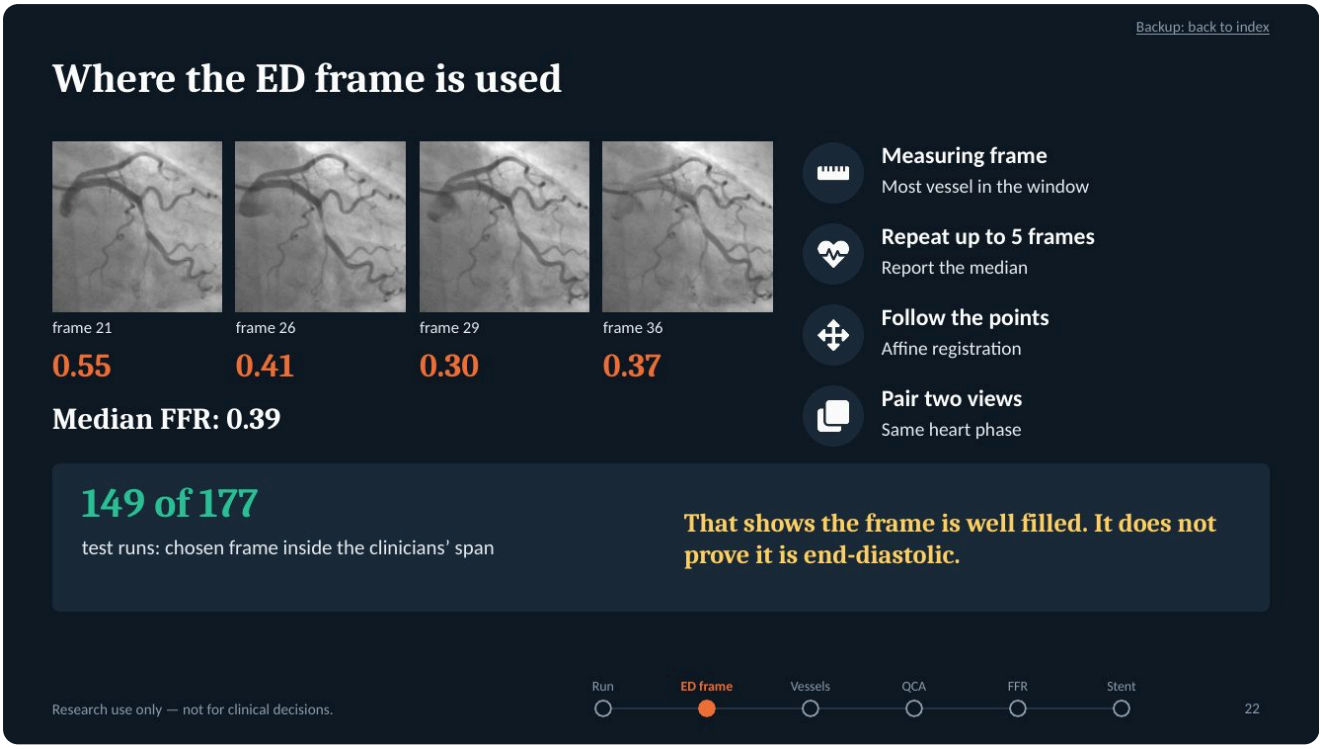
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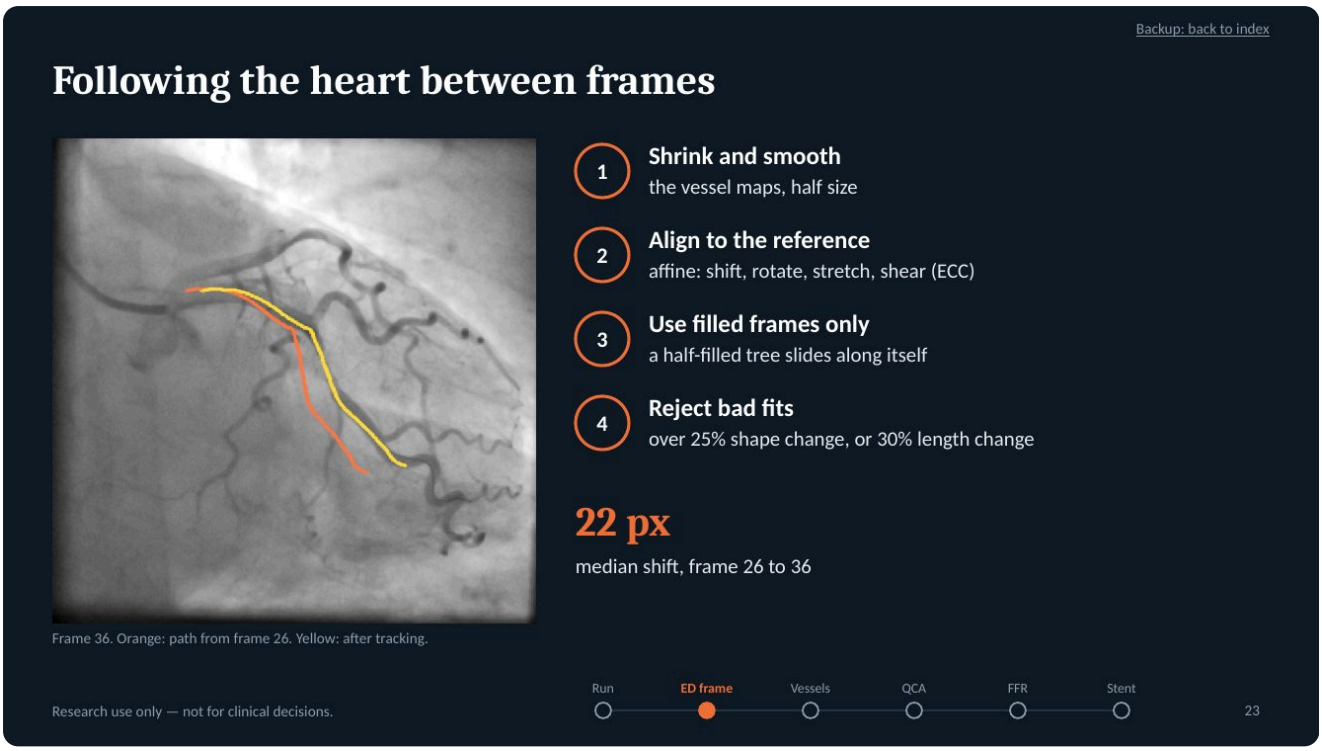
Route two, the ECG. This is a real recording from DEFINE-FLOW. First we keep only the 5 to 15 hertz band, where the sharp QRS spike lives. Then we square the slope and average over 150 milliseconds. That gives the blue energy bumps, one per heartbeat. We pick peaks at least 250 milliseconds apart, refine each to the exact R wave, and take the film frame at that moment. On this one 64-second recording it matched 69 of the 73 stored R-wave markers, within 80 milliseconds. That is one recording, a demonstration only.



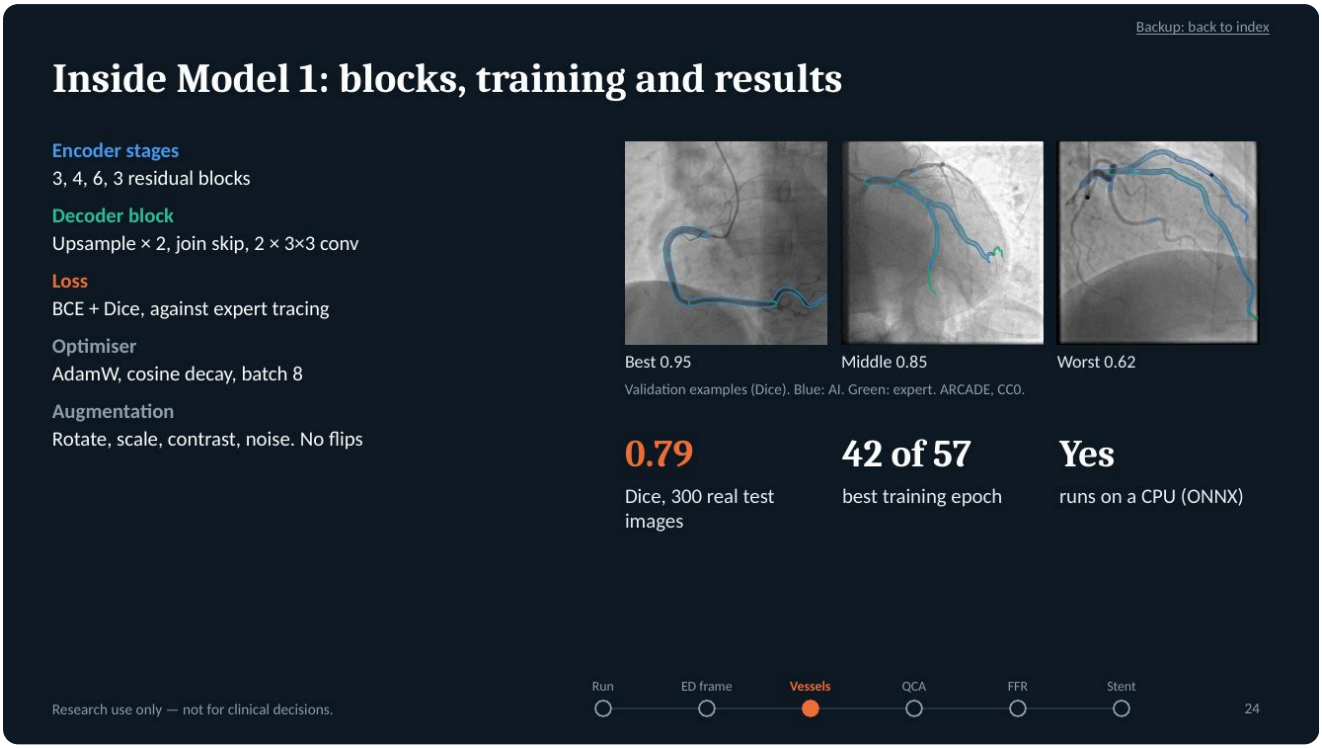
Route three, for files with no ECG. When the heart fills, the vessel tree spreads out. So on each frame we take confident vessel pixels and measure how far they sit from their centre: the spread. We keep well-filled frames, remove the slow trend, and use autocorrelation to find the repeat length of the signal. The peaks, one per beat, are the ED frames. Here is the honest part. On this real run it finds a beat of 3 frames. At about 10 frames per second that is around 200 beats a minute, which is not believable. The curve shows no clear heartbeat. So on CADICA, treat these frames as estimates.



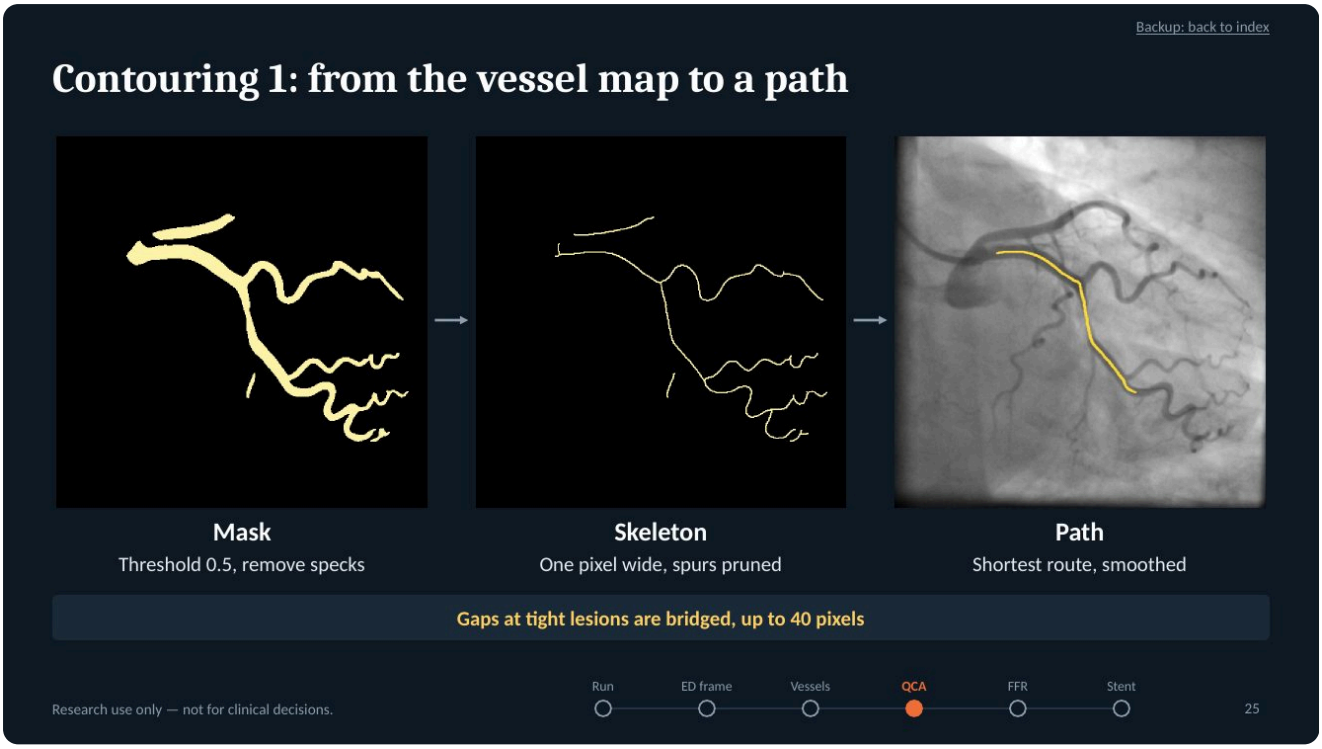
Where is the ED frame used? Four places. First, it picks the analysis frame: the ED frame in the usable window with the most vessel. QCA runs there. Second, we repeat on other ED frames where the path is fully filled, up to five, and report the median. Here four frames gave 0.55, 0.41, 0.30 and 0.37, so the median is 0.39. Third, motion tracking: a registration carries our two clicked points to each frame. Fourth, two-view 3D pairs views at the same heart phase. How do we check it on real data? Only indirectly: the chosen frame fell inside the clinicians' span in 149 of 177 test runs. That shows it is well filled. It does not prove it is end-diastolic.



The heart moves, so a point clicked on one frame is in the wrong place on the next. This picture shows it. Orange is the path from frame 26, drawn on frame 36. Yellow is the same path after tracking, and it sits back on the vessel. The shift here is about 22 pixels. How? We shrink each vessel map to half size, smooth it, and align it to the reference frame with an affine transform: shift, rotate, stretch and shear. The method is called ECC. We only use well-filled frames, because a half-filled tree slides along itself. A fit that changes shape by more than 25 percent is rejected. If a tracked path changes length by more than 30 percent, it has probably jumped to another vessel, so that frame is left out.

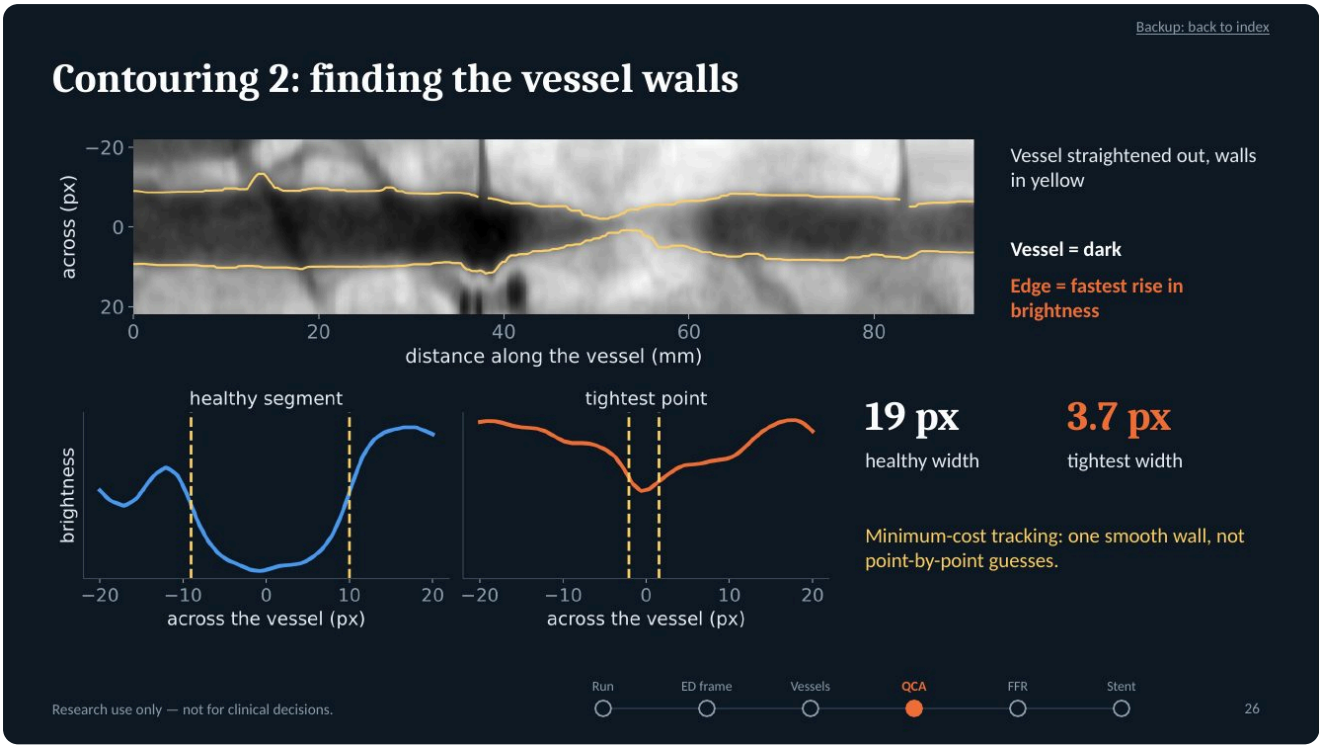


Inside, the encoder uses residual blocks: 3, 4, 6 and 3 of them, in four stages. Each decoder block doubles the size, joins the skip connection, and applies two 3-by-3 convolutions. Training works like this. The network sees a picture, guesses, and the BCE plus Dice loss scores the guess against the expert tracing. AdamW adjusts the weights. We rotated and brightened images for variety, but never flipped them, because left and right matter in anatomy. Best epoch was 42 of 57. On 300 real test images the mean Dice is 0.79. These three examples show best, middle and worst, and the worst one shows a vessel the experts did not trace.



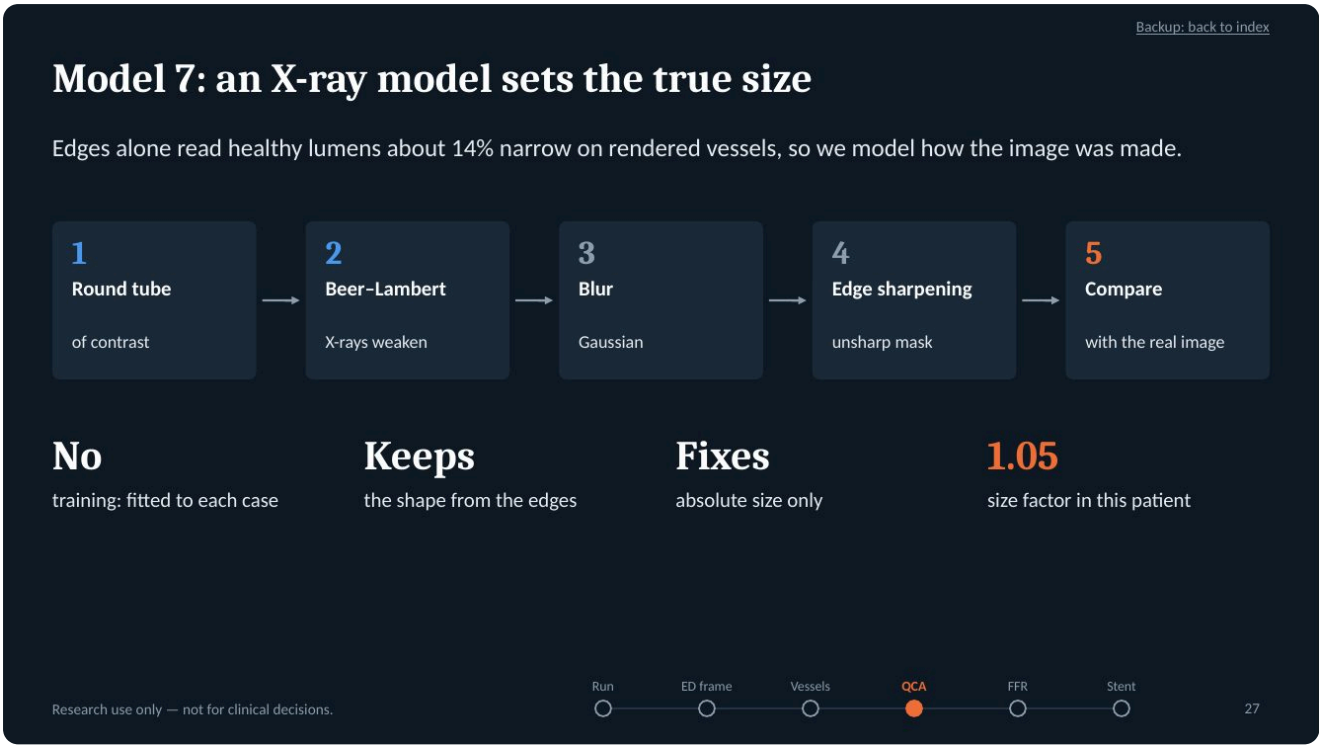
Contouring has two stages. First, from the network's map to a path. We threshold at 0.5 to get a mask and remove specks. We thin it to a one-pixel skeleton, then turn that into a graph of connected pixels, and prune small spurs under 10 pixels. Networks often break a vessel exactly at a tight lesion, so we bridge gaps of up to 40 pixels when two vessel ends face each other. Then the computer finds the shortest route along the graph between the start and end points, and smooths it. The user can click two points, or the software follows the main trunk automatically, up to 100 millimetres.

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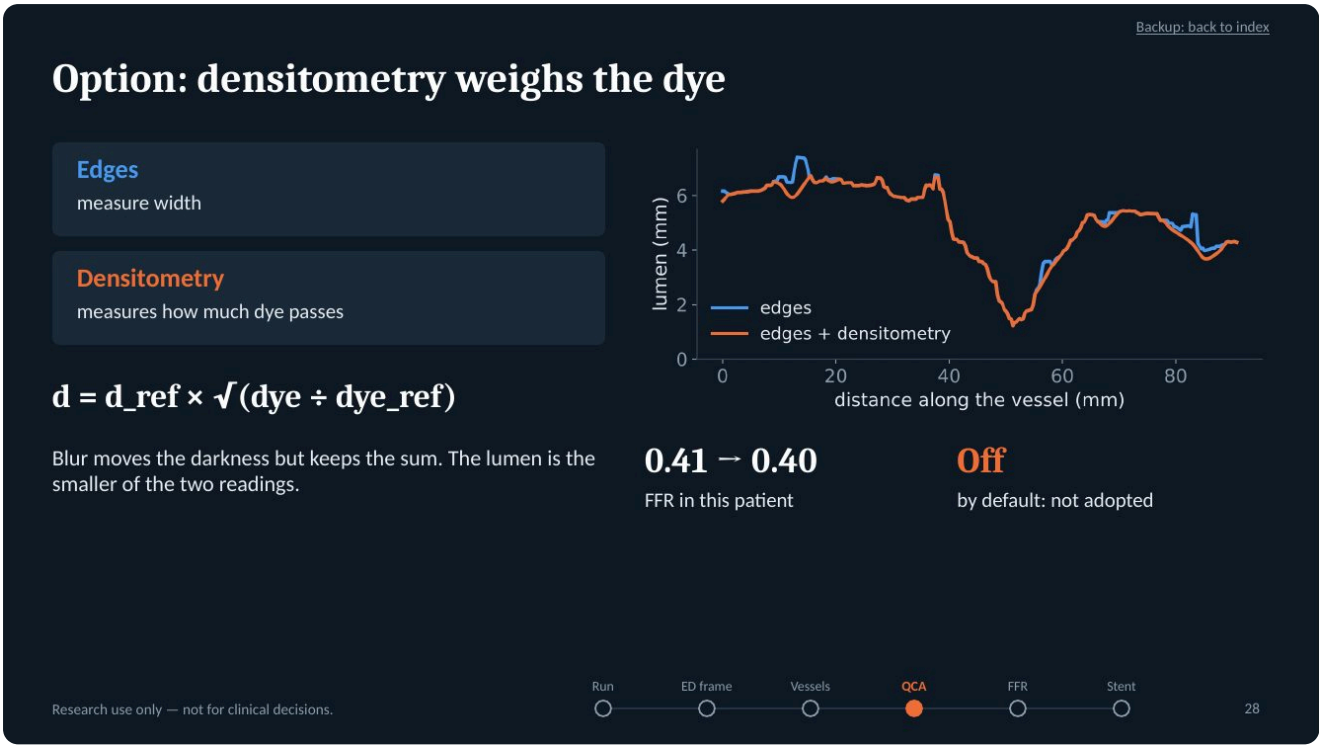


Second stage: the contour itself. At every point on the path we walk straight across the vessel, in steps of a tenth of a pixel, and read brightness. The vessel is dark, so an edge is where brightness rises fastest. The top picture is the vessel straightened out, with both contours in yellow. A wall is continuous, so we do not choose each edge alone. We find the single cheapest contour along the whole vessel, a method called minimum-cost tracking. Then we smooth over 5 points. Look at the bottom left: healthy vessel, a deep valley 19 pixels wide. Bottom right: the tightest point, a shallow dip only 3.7 pixels wide. That is why tight lesions are hard: the dip is near the limit of the blur. Near side branches we keep the network's outline instead.

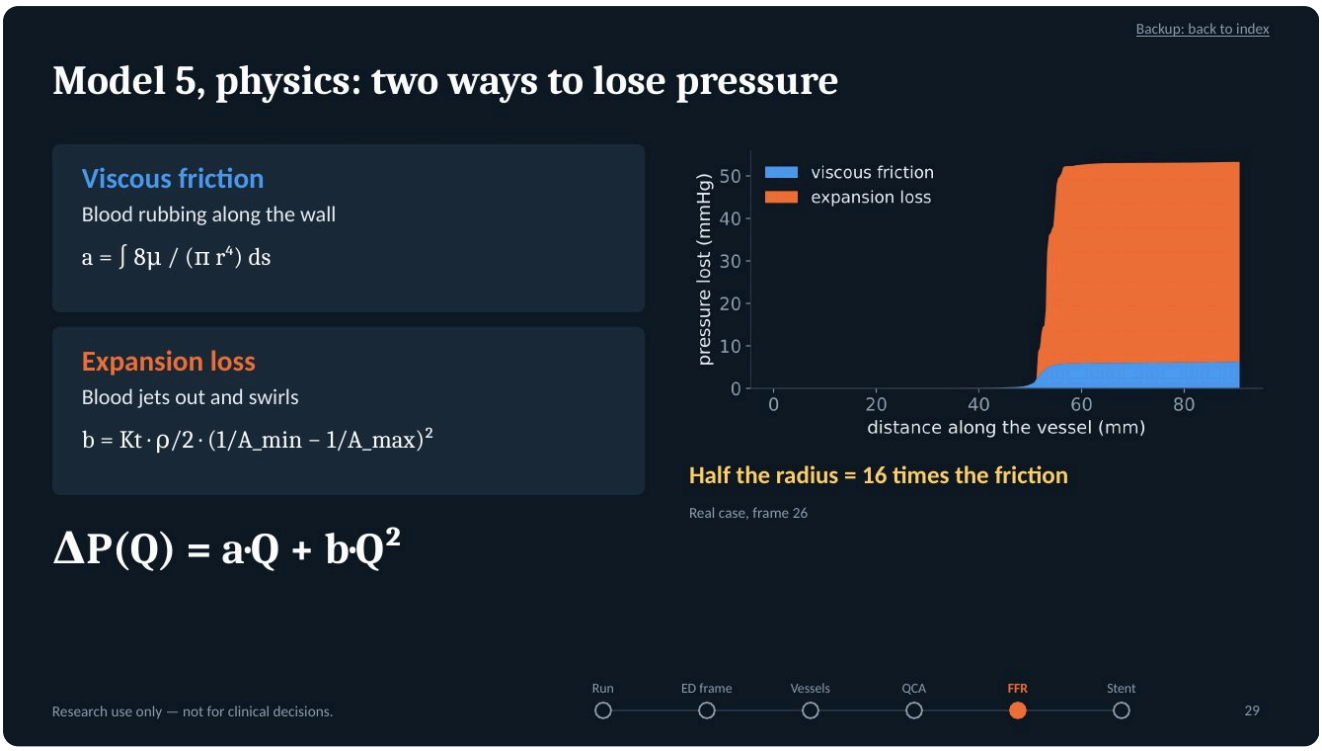
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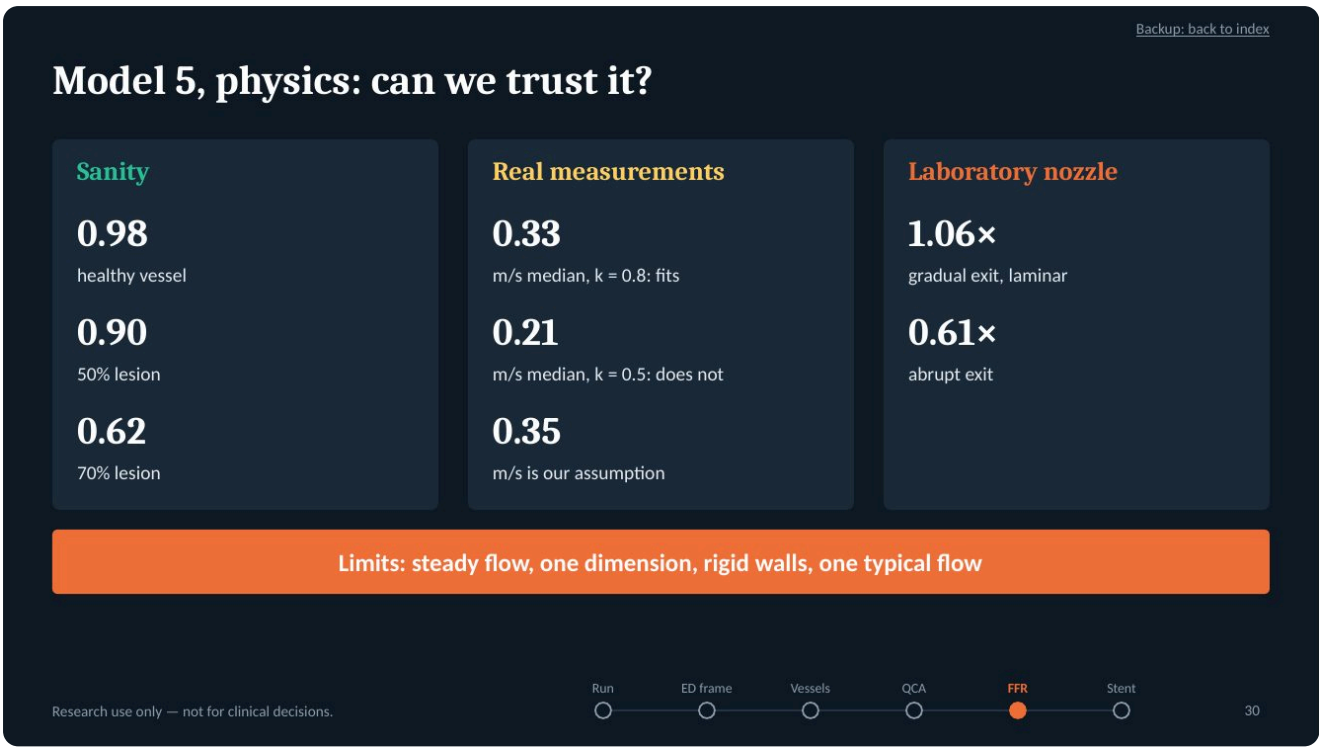
Edges alone have a problem: blur and image processing shift them, so healthy lumens read about 14 percent too narrow on rendered vessels. So we fit a small model of the X-ray system. Imagine a round tube of contrast. X-rays pass through it and weaken, following Beer-Lambert. The system blurs the result, then sharpens edges. We fit this chain to the healthy part of the vessel, and use it to set the true size. It has no training. It keeps the shape from the edges and corrects only the absolute size, by one factor per vessel. Here the factor is 1.05.



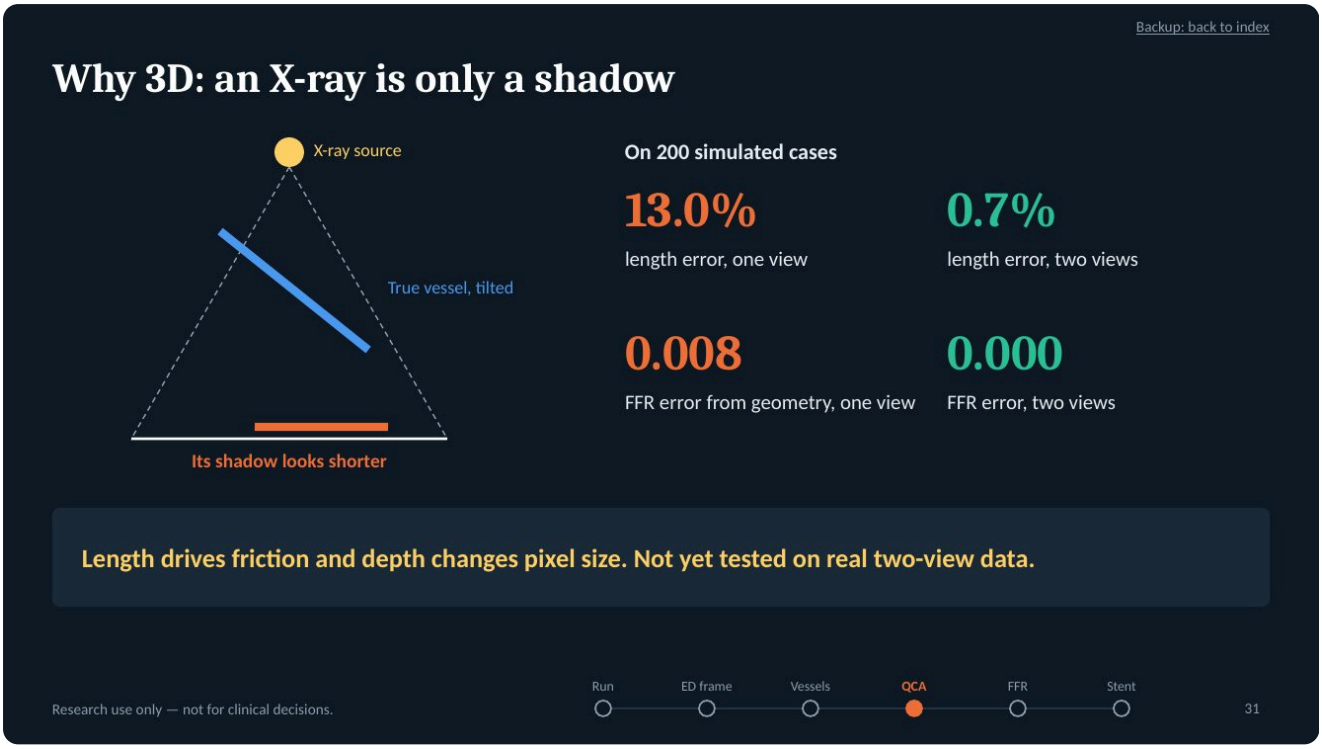
Densitometry is an optional method. Edges measure width. Densitometry measures how much dye is there. Darkness along an X-ray depends on how much dye the ray crosses, so summing darkness across the vessel gives the lumen area. Blur moves darkness around, but does not change the sum. That is the appeal. Diameter is the reference diameter times the square root of the area ratio. The lumen is then the smaller of the edge reading and this reading. In this patient it barely changes anything: 0.41 becomes 0.40. Across all test lesions it helped but did not fix the problem, as the results will show, so it stays switched off by default.



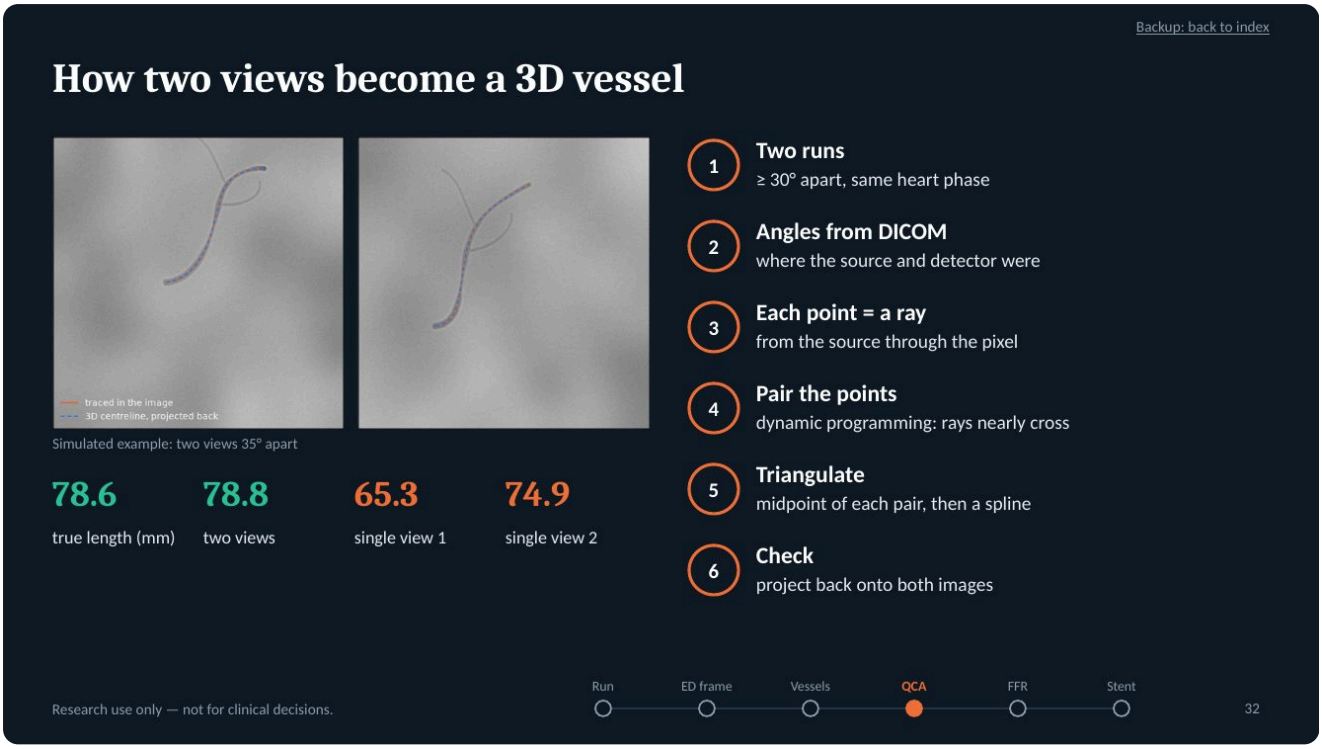
Now the physics, step by step. There is no AI here. A narrowing costs pressure in two ways. First, viscous friction: blood rubbing along the wall. It grows with length and with one over radius to the fourth, so halving the radius multiplies it by 16. Second, expansion loss: blood jets out of the narrowing and swirls, wasting energy. The constant 1.52 comes from published experiments. The chart shows both along our real vessel. Flat in the healthy part, then a sudden jump. Most of the loss here is expansion, not friction.



Do we trust the physics? We tested it three ways. Basic sense: a healthy vessel gives 0.98, a 50 percent lesion 0.90, a 70 percent lesion 0.62. Real measurements: in DEFINE-FLOW, our assumed 0.35 metres per second sits inside the measured range for the more realistic Doppler factor, median 0.33, but above it for the other, median 0.21. So flow is a real error source. Laboratory: on the FDA nozzle, a gradual exit matches at 1.06 times the measured loss at laminar flow, but an abrupt exit gives only 0.61 times. And the model is simple: steady flow, one dimension, rigid walls, one typical flow.

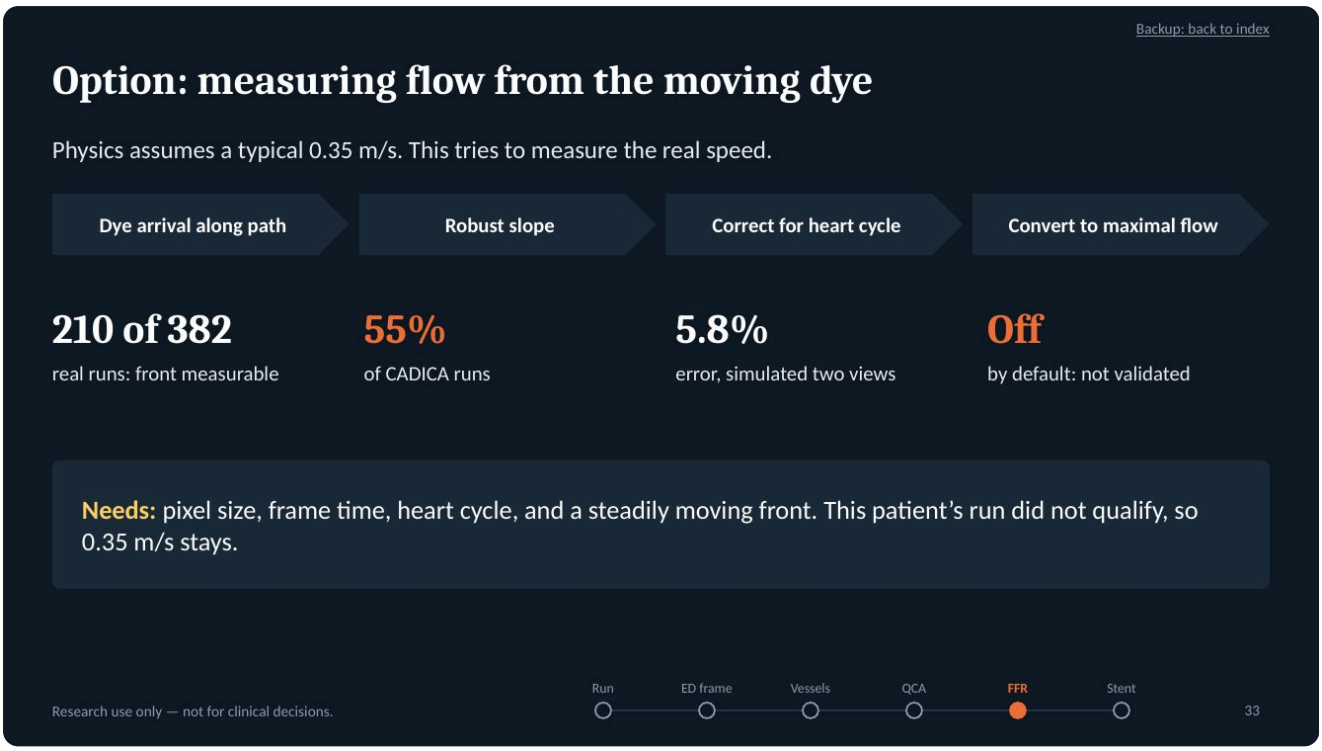


Why reconstruct in 3D? Because an X-ray is a shadow. A vessel tilted towards the beam looks shorter than it is. That is foreshortening. And the pixel size depends on how deep the vessel lies, so one view gets both length and size slightly wrong. This matters because friction grows with length, and depends hugely on radius. On 200 simulated cases, one view misjudged length by a median of 13 percent. Two views misjudged it by 0.7 percent. The FFR error from geometry fell from 0.008 to almost zero. One warning: these cases are simulated, and they are the method's best case. CADICA has no angles, so it has not been tested on real two-view data.

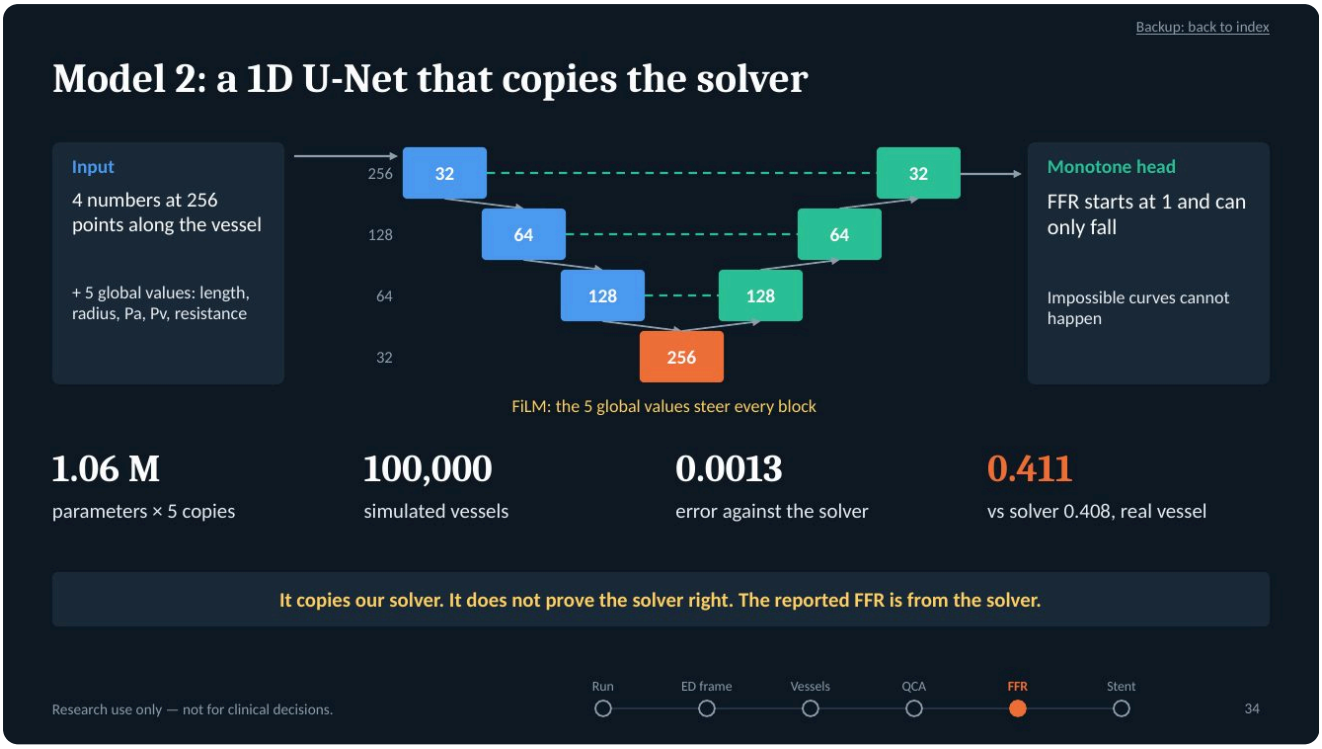


How does it work? We need two runs of the same vessel, at least 30 degrees apart, at the same heart phase, so ED frames matter here. The DICOM file gives the X-ray angles and distances. Each point on a 2D centreline defines a ray from the X-ray source. A true pair of points has rays that nearly cross. Dynamic programming finds the best pairing from start to end. The midpoint where each pair of rays comes closest is the 3D point. A smooth spline runs through those points. Then we check our work: project the 3D line back onto both images and see if it lands on the vessels. This example is simulated: true length 78.6 millimetres, two views 78.8, single views 65.3 and 74.9.

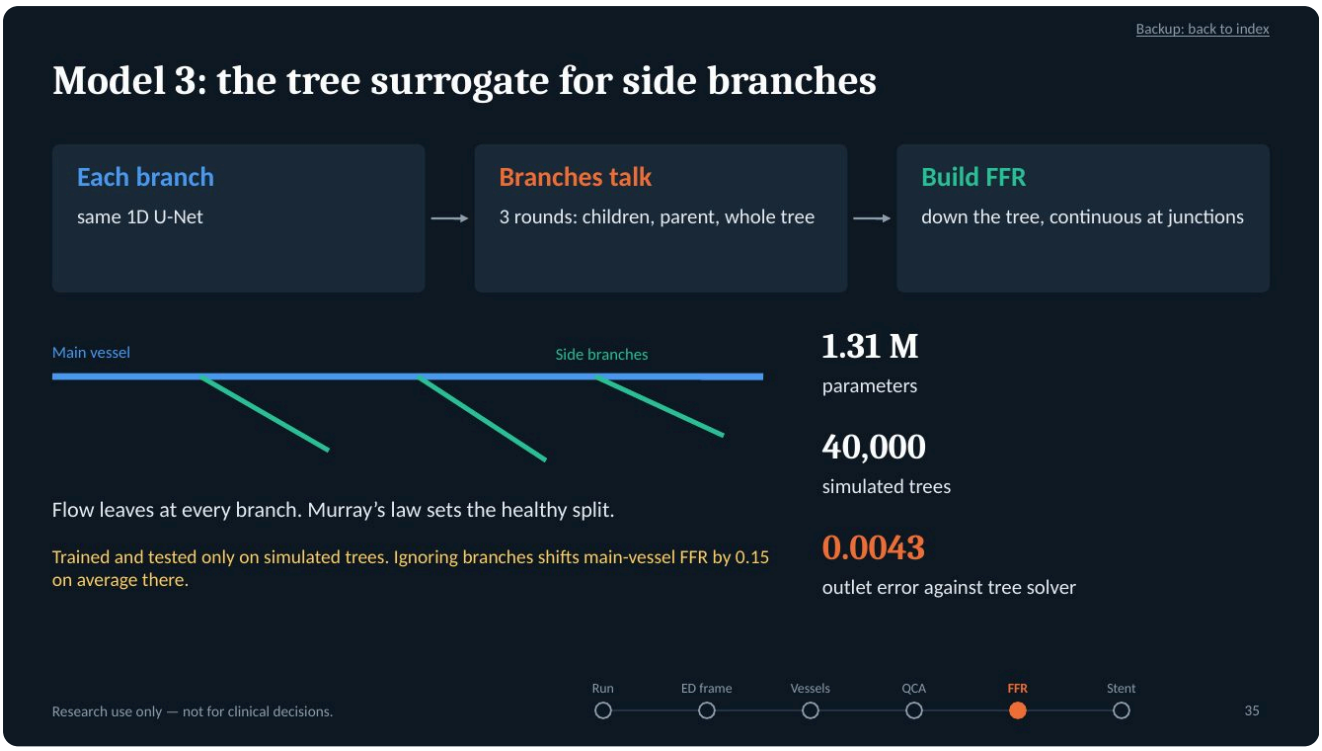
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Our physics assumes a typical flow, 0.35 metres per second. Can we measure it instead? We can try, from how fast the dye front moves along the vessel. We take the arrival times, and fit a robust slope of distance against time. Flow is faster in diastole than systole, so we correct using the ED frames. A resting run is converted to maximal flow with a published formula. This is off by default. It needs pixel size and frame time, which CADICA lacks. On simulated two-view runs the median error was 5.8 percent. On real CADICA runs, a steady moving front could be measured in 210 of 382 runs, 55 percent, and this patient's run was not one of them. Since it needs validation, the fixed value stays.

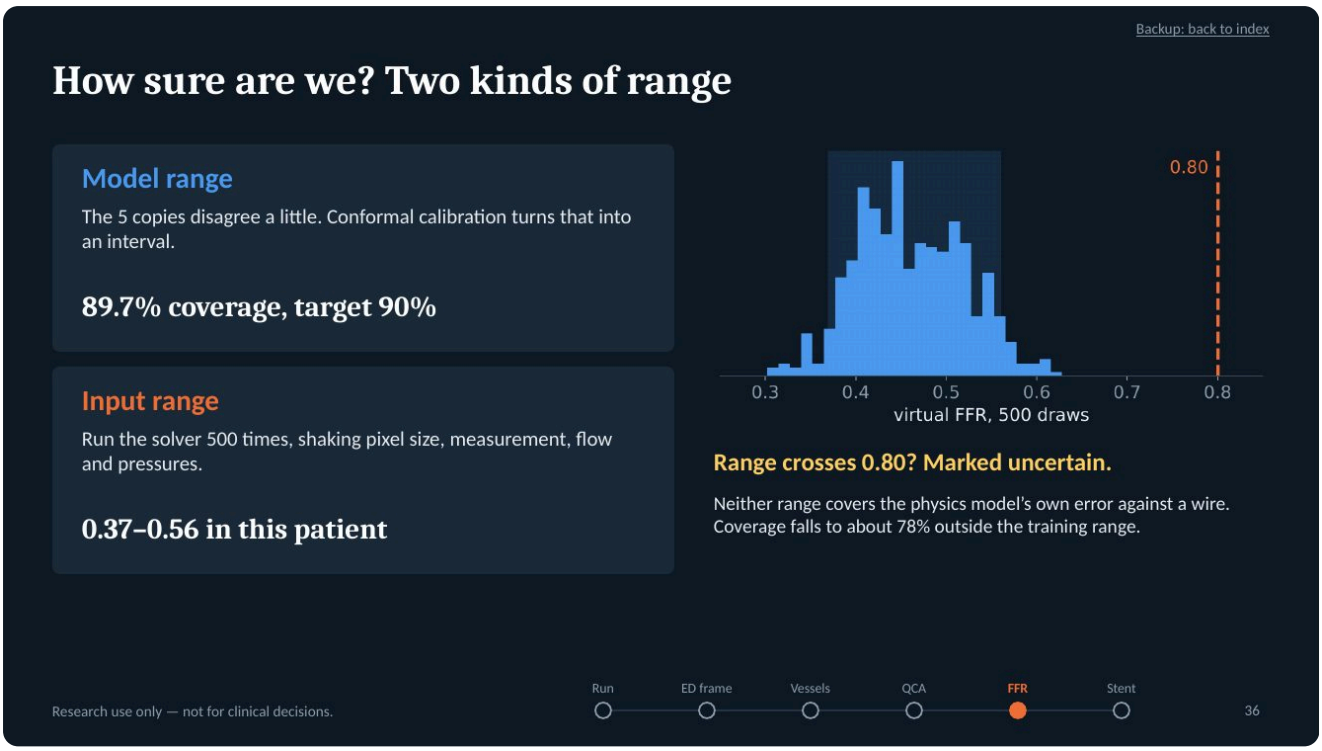


Model two imitates the physics solver. It is also a U-Net, but one-dimensional: it slides along the vessel, not across a picture. Input is four numbers at 256 points along the vessel. Five global values, such as length and pressures, steer each block, a method called FiLM. The monotone head forces FFR to start at 1 and only fall, so impossible curves cannot happen. We trained five copies on 100,000 simulated vessels. Its error against the solver is 0.0013. Model 4, a simple gradient-boosting baseline, had 0.034, about 26 times worse. On our real vessel it gave 0.411 against the solver's 0.408. Remember: it copies our solver, it does not prove the solver is right. And the reported FFR still comes from the solver.



Model three handles side branches, because flow leaves the main vessel at each branch. Every branch goes through the same kind of 1D U-Net. Then branches talk to each other: three rounds of message passing, from children, parent and the whole tree. That matters because flow in one branch depends on all others. The head builds FFR down the tree, so pressure is continuous at each junction. It has 1.31 million parameters and was trained on 40,000 simulated trees, with outlet error 0.0043 against the tree solver. The tree solver itself splits flow at junctions using Murray's law. Ignoring branches changed main-vessel FFR by 0.15 on average on simulated trees, so branches do matter.

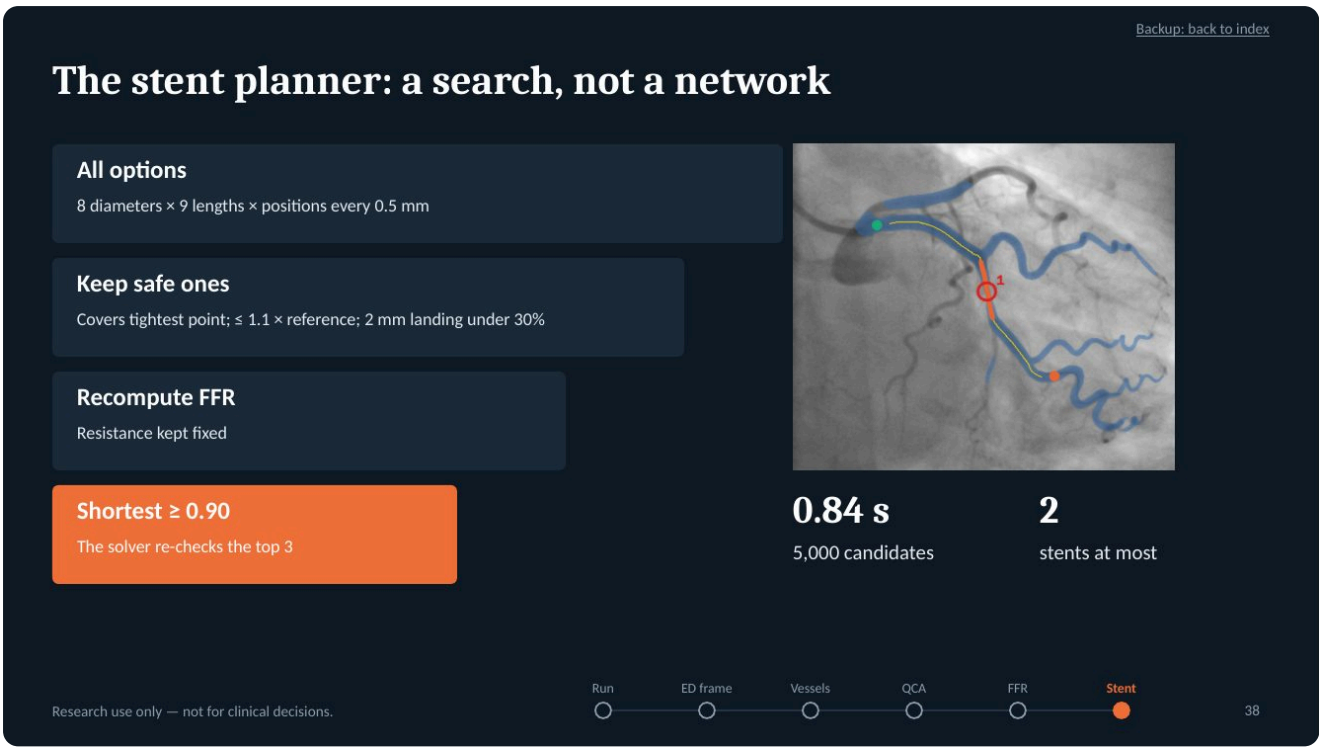
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A single number is not enough, so each FFR carries a range, from two sources. First, model error. The five copies disagree a little, and conformal calibration turns that into an interval: 89.7 percent coverage for a 90 percent target on simulated vessels, and only about 78 percent outside the training range. Second, input error. We re-run the solver 500 times, shaking pixel size, measurement error, flow and pressures. For this patient that gives 0.37 to 0.56. Measurement error and flow matter most. Because the whole range is below 0.80, it is not flagged. If it crossed 0.80 it would be marked uncertain. Neither range covers the physics model's own error against a real wire.

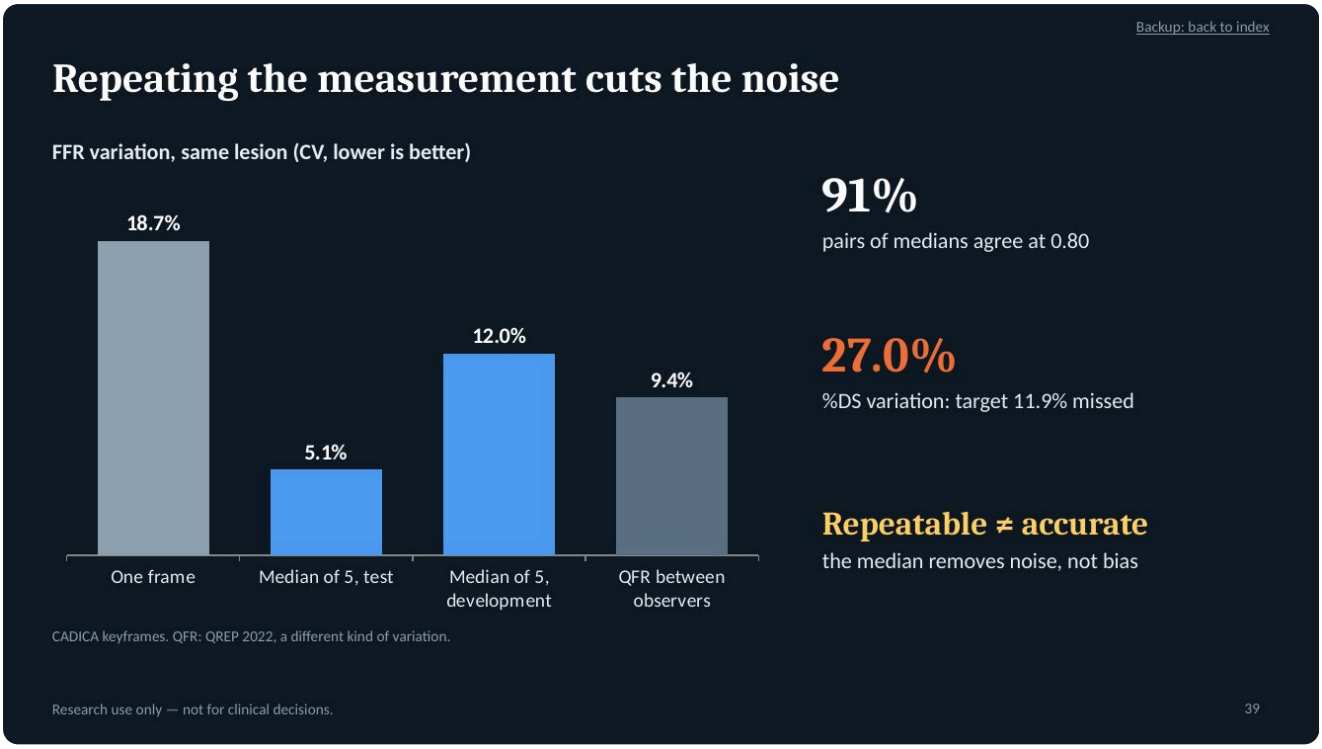


What does a virtual stent do to the vessel? It is simple geometry. Inside the stent, the lumen opens to the stent's radius. If the vessel is already wider, nothing changes: a stent never narrows the lumen. The last millimetre at each end blends smoothly. The chart shows the real vessel: blue now, green with the stent. Then we rerun the physics with the new lumen. One important choice: small-vessel resistance stays at its value before the stent, because a stent changes the artery, not the heart muscle's small vessels. So flow rises, and FFR recovers from 0.41 to 0.96.



The planner is a search, not a neural network. The generic catalogue has 8 diameters from 2.25 to 4.00 millimetres and 9 lengths from 8 to 38. It tries every combination at every position, in half-millimetre steps. Then it applies safety rules. The stent must cover the tightest point. It must not exceed 1.1 times the reference diameter. The first and last 2 millimetres must land where narrowing is under 30 percent. For each plan left, it recomputes FFR. It chooses the shortest stent reaching 0.90, and if none does, the highest FFR. Two lesions can have two stents, from the best 20 options for each. The surrogate screens quickly, 5,000 candidates in 0.84 seconds, then the solver re-checks the top three. On 100 simulated vessels the target was reached in 84.

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First group result, on real angiograms: does the number repeat? On a single frame, FFR for the same lesion varies by 18.7 percent. That is not good enough. The median of five frames gives 5.1 percent on test patients, with an interval of 3.4 to 6.8. On development patients the same rule gave 12.0, so the truth may lie between. QFR's 9.4 percent between observers is shown for scale, though it measures something different. Percent narrowing still varies by 27 percent, missing the 11.9 target. And repeatable does not mean correct: a median removes noise, not a bias shared by every frame. These tests used CADICA keyframes, not ED frames.

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